

Multi-Crystal Arrays

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11th December 2003

Abstract

It is possible to treat multi-crystal arrays on a crystal by crystal basis. This requires making adjustments to the standard transmission probability functions for resolution calculations.

transmission probability functions

Now let us look at the transmission probability function (TPF) for a three axis neutron spectrometer[CandN]. The transmission function of a crystal element is

$$P_X = N_X \exp \left(-\frac{1}{2} \left\{ \left(\frac{\gamma_i + \gamma_f}{2\eta_h} \right)^2 + \left(\frac{\delta_f - \delta_i}{2\eta_v \sin(\theta_X)} \right)^2 \right\} \right) \quad (1)$$

where γ_i and γ_f are the deviation angles with respect to optimum before and after the crystal in the scattering plane, and δ_i and δ_f are the corresponding deviation angles out of the scattering plane. The horizontal and vertical mosaics of the crystal are η_h and η_v and θ_X is the Bragg angle for the reflection. The normalization, N_X , is given by

$$N_X = R_X / \left[\sqrt{2\pi} 2\eta_{Xv} \sin(\theta_X) \right] \quad (2)$$

where R_X is the crystal reflectivity in the range zero to one, and the remaining factor is the vertical deviation angle normalization, since the total probability of transmission does not depend on the vertical mosaic. The vertical mosaic simply contributes to the vertical spread of the beam of neutrons

reflected. Thus we have the horizontal part of the transmission function from the reactor up to the sample (leaving out the normalization) as,

$$P_{Mh} = \exp \left(-\frac{1}{2} \left\{ \left(\frac{\gamma_0}{\alpha_0} \right)^2 + \left(\frac{\gamma_0 + \gamma_1}{2\eta_{Mh}} \right)^2 + \left(\frac{\gamma_1}{\alpha_1} \right)^2 \right\} \right) \quad (3)$$

where γ_0 and γ_1 are the horizontal deviation angles of the neutron from the nominal path before and after the monochromator with positive angles c.c.w., α_0 and α_1 are the corresponding Soller collimation standard deviations, and η_{Mh} is the horizontal mosaic standard deviation for the monochromator. Similarly the vertical part of this transmission function is

$$P_{Mv} = N_M \exp \left(-\frac{1}{2} \left\{ \left(\frac{\delta_0}{\beta_0} \right)^2 + \left(\frac{\delta_1 - \delta_0}{2\eta_{Mv} \sin(\theta_M)} \right)^2 + \left(\frac{\delta_1}{\beta_1} \right)^2 \right\} \right) \quad (4)$$

where δ_0 and δ_1 are the vertical deviation angles of the neutron from the nominal path before and after the monochromator with positive angles up, β_0 and β_1 are the corresponding Soller collimation standard deviations, and η_{Mv} is the vertical mosaic standard deviation for the monochromator.

The Bragg reflection of the neutron correlates the horizontal deviation angles to the wavelength so that

$$\gamma_1 = \gamma_0 - 2 \frac{\Delta k_i}{k_I} \tan(\omega_M) \quad (5)$$

where

$$\Delta k_i = k_i - k_I \quad (6)$$

and

$$k_I = \frac{\pi}{d_M \sin(\omega_M)} \quad (7)$$

Here ω_M is the signed angle of the reflection plane with respect to the nominal incident path and this means that γ_1 is measured with respect to the outgoing path at $2\omega_M$. The sign of ω_M handles both left and right scattering again with positive c.c.w.

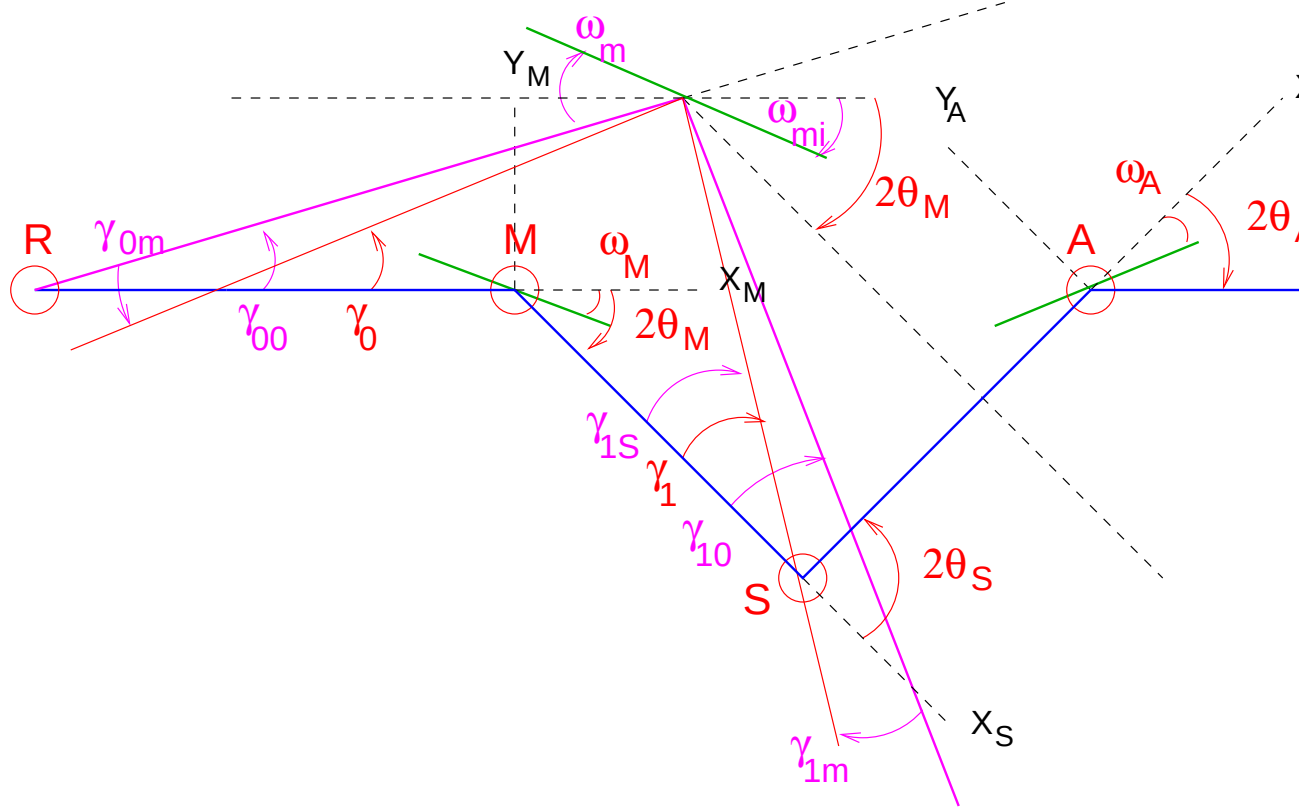
transmission probability for horizontally displaced crystals

In order to account for the effects on the TPF of horizontally displaced monochromator crystals, we must add the following terms to the sum inside the exponential in equation 3,

$$\left(\frac{\gamma_0 - \gamma_{00}}{\alpha_{RM}}\right)^2 + \left(\frac{\gamma_1 - \gamma_{1S}}{\alpha_{MS}}\right)^2 + \left(\frac{\gamma_{10} - \gamma_{1S}}{\alpha_{MS}}\right)^2 \quad (8)$$

where α_{RM} and α_{MS} are the geometrical collimations from the reactor to monochromator, and monochromator to sample, due to the sizes of these elements. Note that the same types of terms would describe horizontal radial collimation, but with α_{RM} and/or α_{MS} replaced by the value of the standard deviation angle for the radial collimator.

The following diagram shows the angles involved when the monochromator is displaced horizontally.



As above, γ_{00} is the horizontal deviation angle of the crystal element with respect to the nominal reactor to monochromator line, and γ_{10} is the corresponding deviation angle of the nominal outgoing beam path from the crystal element to the sample. The outgoing path that corresponds to the actual crystal tilt is γ_{10} , while γ_{1S} is for the path that is directed at the sample. Ideally we would like $\gamma_{1S} = \gamma_{10}$ so that the crystal is set to point the nominal outgoing direction at the sample position. γ_{0m} is the neutron path deviation from the γ_{00} line, and γ_{1m} is the neutron path deviation from the nominal outgoing path of the crystal element, while as usual γ_0 and γ_1 are the neutron path deviation angles with respect to the original reactor to monochromator line and monochromator to sample line respectively. The new crystal rotation angle, ω_{mi} , is still measured with respect to the original reactor to monochromator line, however the effective reflection angle, ω_m , has to be measured with respect to the new nominal path to the displaced crystal. Then we find that

$$\omega_{mi} = \omega_m + \gamma_{00} \quad (9)$$

$$\omega_{mi} + \omega_m = 2\theta_M + \gamma_{10} \quad (10)$$

so that

$$\gamma_{10} = \gamma_{00} + 2(\omega_m - \omega_M) \quad (11)$$

It is still true for deviation pathes with respect to the new nominal path to the displaced crystal, that

$$\gamma_{1m} = \gamma_{0m} - 2\frac{\Delta k_{im}}{k_{Im}} \tan(\omega_m) \quad (12)$$

where

$$\Delta k_{im} = k_{im} - k_{Im} \quad (13)$$

and

$$k_{Im} = \frac{\pi}{d_M \sin(\omega_m)} \quad (14)$$

Note also that

$$\omega_{mi} - \gamma_0 = \omega_m - \gamma_{0m} \quad (15)$$

Also we want γ_1 and γ_0 in terms of γ_{1m} and we find that

$$\gamma_1 = \gamma_{1m} + \gamma_{10} \quad (16)$$

$$\gamma_0 = \gamma_{0m} + \gamma_{00} = \gamma_{1m} + 2\frac{\Delta k_{im}}{k_{Im}} \tan(\omega_m) + \gamma_{00} \quad (17)$$

Now the correlation between the original deviation angles becomes

$$\gamma_1 = \gamma_0 - 2\frac{\Delta k_{im}}{k_{Im}} \tan(\omega_m) + 2(\omega_m - \omega_M) \quad (18)$$

However, we would like to write the transmission function in terms of the displaced pathes using

$$\gamma_0 - \gamma_{00} = \gamma_{0m} \quad (19)$$

$$\gamma_1 - \gamma_{10} = \gamma_{1m} \quad (20)$$

$$\gamma_{1S} - \gamma_{10} = \gamma_{MS} \quad (21)$$

Thus

$$P_{Mh} = \exp\left(-\frac{1}{2}A_{Mh}^2\right) \quad (22)$$

where

$$A_{Mh}^2 = \left(\frac{\gamma_0}{\alpha_0}\right)^2 + \left(\frac{\gamma_{0m} + \gamma_{1m}}{2\eta_M}\right)^2 + \left(\frac{\gamma_1}{\alpha_1}\right)^2 + \left(\frac{\gamma_{0m}}{\alpha_{RM}}\right)^2 + \left(\frac{\gamma_{1m}}{\alpha_{MS}}\right)^2 + \left(\frac{\gamma_{MS}}{\alpha_{MS}}\right)^2 \quad (23)$$

or

$$A_{Mh}^2 = \left(\frac{\gamma_1 + 2\overline{\Delta k_{im}} \tan(\omega_m) - 2\Delta\omega_m}{\alpha_0}\right)^2 + \left(\frac{\gamma_1 + \overline{\Delta k_{im}} \tan(\omega_m) - \gamma_{10}}{\eta_M}\right)^2 + \left(\frac{\gamma_1}{\alpha_1}\right)^2 + \left(\frac{\gamma_1 + 2\overline{\Delta k_{im}} \tan(\omega_m) - \gamma_{10}}{\alpha_{RM}}\right)^2 + \left(\frac{\gamma_1 - \gamma_{10}}{\alpha_{MS}}\right)^2 + \left(\frac{\gamma_{MS}}{\alpha_{RM}}\right)^2 \quad (24)$$

and we have defined

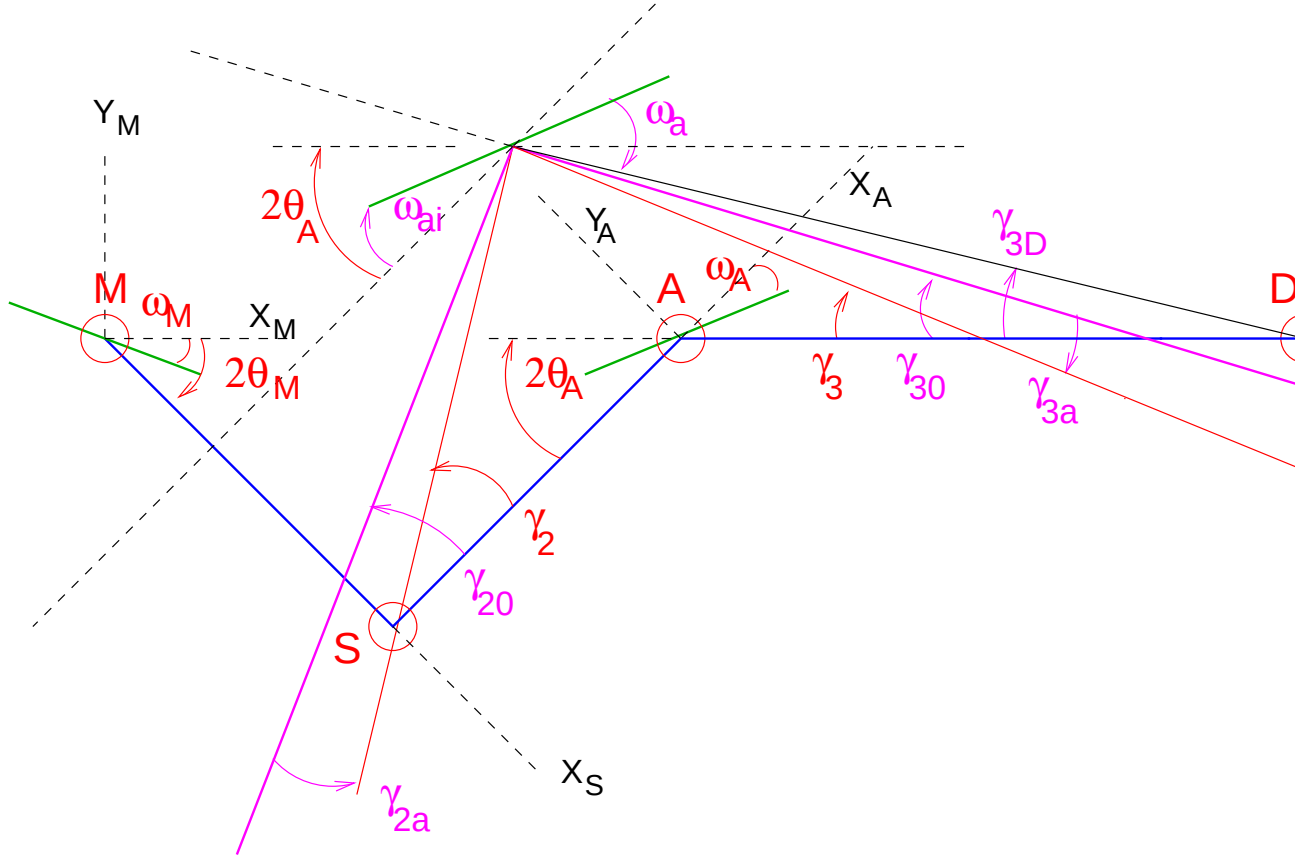
$$\overline{\Delta k_{im}} = \frac{\Delta k_{im}}{k_{Im}} \quad (25)$$

and

$$\Delta\omega_m = \omega_m - \omega_M \quad (26)$$

For now we shall let $\gamma_{MS} = 0$ or $\gamma_{1S} = \gamma_{10}$.

The next figure shows the angles for a horizontally displaced analyzer element.



Here we assume that there are no losses when $\gamma_{30} \neq \gamma_{3D}$, as would be the case for a position sensitive detector. Using a PSD requires identifying the counts associated with a monochromator-analyzer crystal pair. The spot on the detector for this pair can be calculated from the analyzer position and angle. Taking $\mathbf{r}_A$ as the displacement of the analyzer, and including a

vertical path offset angle, δ_{30} , it is easy to show that the detector coordinates of this beam are

$$y_D = \mathbf{r}_A \bullet \hat{\mathbf{y}}_D + (L_{AD} - \mathbf{r}_A \bullet \hat{\mathbf{x}}_D) \tan(\gamma_{30}) \quad (27)$$

$$z_D = \mathbf{r}_A \bullet \hat{\mathbf{z}} + (L_{AD} - \mathbf{r}_A \bullet \hat{\mathbf{x}}_D) \tan(\delta_{30}) / \cos(\gamma_{30}) \quad (28)$$

where L_{AD} is just the nominal distance from the original analyzer origin to the detector origin.

Thus the horizontal transmission function after the sample is

$$P_{Ah} = \exp\left(-\frac{1}{2}A_{Ah}^2\right) \quad (29)$$

where

$$A_{Ah}^2 = \left(\frac{\gamma_2}{\alpha_2}\right)^2 + \left(\frac{\gamma_{2a} + \gamma_{3a}}{2\eta_A}\right)^2 + \left(\frac{\gamma_3}{\alpha_3}\right)^2 + \left(\frac{\gamma_2 - \gamma_{20}}{\alpha_{SA}}\right)^2 + \left(\frac{\gamma_3 - \gamma_{30}}{\alpha_{AD}}\right)^2 \quad (30)$$

and corresponding to equation 18

$$\gamma_3 = \gamma_2 - 2\frac{\Delta k_{fa}}{k_{Fa}} \tan(\omega_a) + 2(\omega_a - \omega_A) \quad (31)$$

Also as before

$$\gamma_2 - \gamma_{20} = \gamma_{2a} \quad (32)$$

$$\gamma_3 - \gamma_{30} = \gamma_{3a} \quad (33)$$

$$\gamma_{20} = \omega_{ai} - \omega_a \quad (34)$$

$$\gamma_{30} = \omega_{ai} + \omega_a - 2\theta_A \quad (35)$$

$$\gamma_{30} = \gamma_{20} + 2(\omega_a - \omega_A) \quad (36)$$

It is still true for deviation pathes with respect to the new nominal path to the displaced crystal, that

$$\gamma_{2a} = \gamma_{3a} + 2 \frac{\Delta k_{fa}}{k_{Fa}} \tan(\omega_a) \quad (37)$$

where

$$\Delta k_{fa} = k_{fa} - k_{Fa} \quad (38)$$

and

$$k_{Fa} = \frac{\pi}{d_A \sin(\omega_a)} \quad (39)$$

Note also that

$$2\omega_A - \omega_{ai} + \gamma_3 = \omega_a + \gamma_{3a} \quad (40)$$

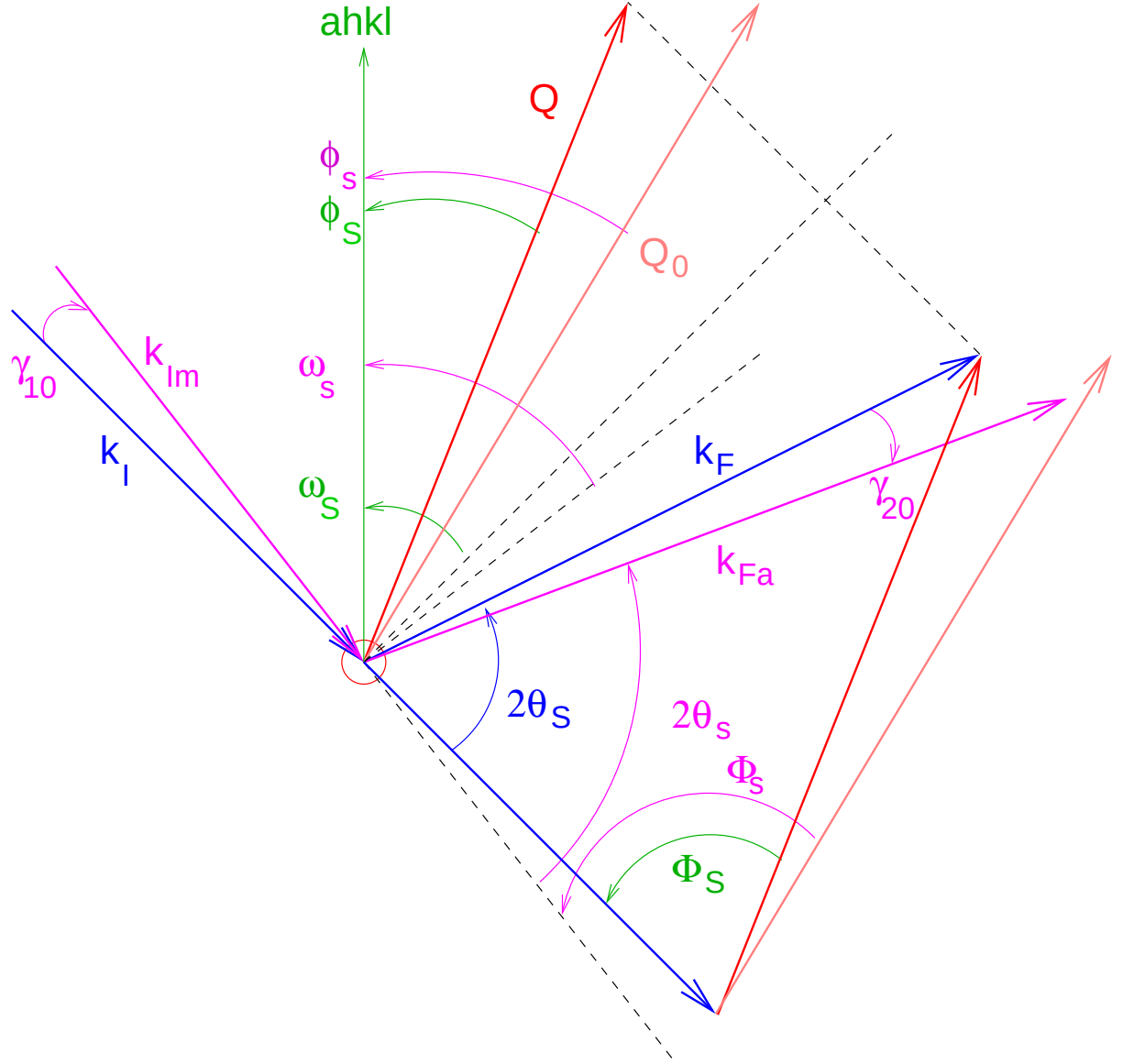
Also we want γ_2 and γ_3 in terms of γ_{2a} and we find that

$$\gamma_2 = \gamma_{2a} + \gamma_{20} \quad (41)$$

$$\gamma_3 = \gamma_{3a} + \gamma_{30} = \gamma_{2a} - 2 \frac{\Delta k_{fa}}{k_{Fa}} \tan(\omega_a) + \gamma_{30} \quad (42)$$

so that we get

$$A_{Ah}^2 = \left(\frac{\gamma_2}{\alpha_2} \right)^2 + \left(\frac{\gamma_2 - \overline{\Delta k_{fa}} \tan(\omega_a) - \gamma_{20}}{\eta_A} \right)^2 + \left(\frac{\gamma_2 - 2\overline{\Delta k_{fa}} \tan(\omega_a) + 2\Delta\omega_a}{\alpha_3} \right)^2 + \left(\frac{\gamma_{2a}}{\alpha_{SA}} \right)^2 + \left(\frac{\gamma_{2a} - 2\overline{\Delta k_{fa}} \tan(\omega_a)}{\alpha_{AD}} \right)^2 \quad (43)$$



The effective wave-vector transfer, Q_0 , from the offset pathes is

$$\mathbf{Q}_0 = \mathbf{k}_{Fa} - \mathbf{k}_{Im} \quad (44)$$

with

$$2\theta_s = 2\theta_S + \gamma_{20} - \gamma_{10} \quad (45)$$

$$\omega_s = \omega_S - \gamma_{10} \quad (46)$$

$$\phi_s = \Phi_s - \Phi_S + \phi_S + \gamma_{10} \quad (47)$$

Recall that

$$Q^2 = k_I^2 + k_F^2 - 2 \cos(2\theta_S) \quad (48)$$

$$\omega_S = \frac{\pi}{2} + \phi_S - \Phi_S \quad (49)$$

$$\tan(\phi_S) = -Q_2/Q_1 \quad (50)$$

$$\cos(\Phi_S) = (k_I^2 - k_F^2 + Q^2) / (2 k_I Q) \quad (51)$$

Also the sample scattering angle is given by

$$\cos(2\theta_S) = (k_I^2 + k_F^2 - Q^2) / (2 k_I k_F) \quad (52)$$

Here Q_1 is the component of $\mathbf{Q}$ along **ahkl** and Q_2 is its component along the effective reflection perpendicular to **ahkl**. There are identical equations with the uppercase S angle subscript replaced by lowercase s, k_I replaced by k_{Im} , k_F replaced by k_{Fa} and Q replaced by Q_0 . Thus we start by calculating Q_0 from equation 48 using k_{Im} , k_{Fa} and $2\theta_s$, then calculate Φ_s and $2\theta_s$ from equations 51 and 52, and then ϕ_s from equation 47 to finally get the Q_{01} and Q_{02} components from equation 50.

vertical displacements and crystal tilts

We also must treat vertical displacements of the crystals including tilts of the crystals about their horizontal axes.

If the monochromator crystal element is displaced vertically so that its vertical deviation angle with respect to the nominal scattering plane is δ_{00} , and rotated about its vertical axis by the angle ω_{mi} with respect to the original line from reactor to monochromator and then rotated by χ_m with respect to the crystal element's horizontal axis, we find that the crystal normal is pointing at

$$\hat{\mathbf{n}}_m = \sin(\chi_m) \hat{\mathbf{z}} + \cos(\chi_m) \left[\frac{\omega_m}{|\omega_m|} \cos(\omega_m) \hat{\mathbf{y}}_m - \sin|\omega_m| \hat{\mathbf{x}}_m \right] \quad (53)$$

where $\hat{\mathbf{x}}_m$ and $\hat{\mathbf{y}}_m$ are the unit vectors along the offset path and perpendicular to it so that

$$\hat{\mathbf{x}}_m = \cos(\gamma_{00}) \hat{\mathbf{x}}_M + \sin(\gamma_{00}) \hat{\mathbf{y}}_M \quad (54)$$

$$\hat{\mathbf{y}}_m = -\sin(\gamma_{00}) \hat{\mathbf{x}}_M + \cos(\gamma_{00}) \hat{\mathbf{y}}_M \quad (55)$$

and so

$$\begin{aligned} \hat{\mathbf{n}}_m = \sin(\chi_m) \hat{\mathbf{z}} - \cos(\chi_m) & \left[\cos(\gamma_{00}) \sin|\omega_m| + \sin(\gamma_{00}) \frac{\omega_m}{|\omega_m|} \cos(\omega_m) \right] \hat{\mathbf{x}}_M + \\ & \cos(\chi_m) \left[\cos(\gamma_{00}) \frac{\omega_m}{|\omega_m|} \cos(\omega_m) - \sin(\gamma_{00}) \sin|\omega_m| \right] \hat{\mathbf{y}}_M \end{aligned} \quad (56)$$

The incident nominal direction upon this crystal element is given by,

$$\hat{\mathbf{i}}_m = \sin(\delta_{00}) \hat{\mathbf{z}} + \cos(\delta_{00}) [\sin(\gamma_{00}) \hat{\mathbf{y}}_M + \cos(\gamma_{00}) \hat{\mathbf{x}}_M] \quad (57)$$

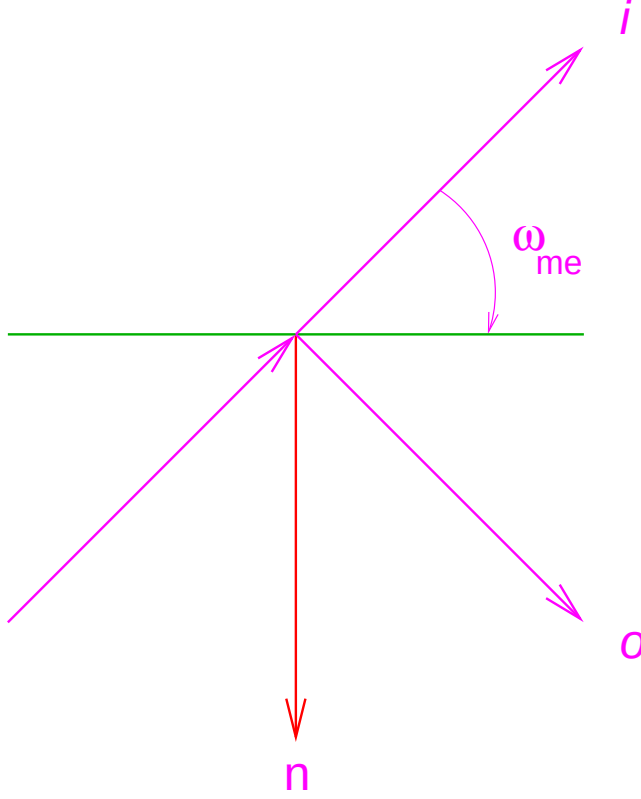
so that

$$\hat{\mathbf{o}}_m = \hat{\mathbf{i}}_m + 2 \sin|\omega_{me}| \hat{\mathbf{n}}_m \quad (58)$$

and

$$\begin{aligned} \hat{\mathbf{o}}_m = [\sin(\delta_{00}) + 2 \sin|\omega_{me}| \sin(\chi_m)] \hat{\mathbf{z}} + \\ \left\{ \cos(\delta_{00}) \sin(\gamma_{00}) + 2 \sin|\omega_{me}| \cos(\chi_m) \left[\cos(\gamma_{00}) \frac{\omega_m}{|\omega_m|} \cos(\omega_m) - \sin(\gamma_{00}) \sin|\omega_m| \right] \right\} \hat{\mathbf{y}}_M + \\ \left\{ \cos(\delta_{00}) \cos(\gamma_{00}) - 2 \sin|\omega_{me}| \cos(\chi_m) \left[\cos(\gamma_{00}) \sin|\omega_m| + \sin(\gamma_{00}) \frac{\omega_m}{|\omega_m|} \cos(\omega_m) \right] \right\} \hat{\mathbf{x}}_M \end{aligned} \quad (59)$$

Referring to the following diagram



$$\begin{aligned}
 -\hat{\mathbf{i}}_m \bullet \hat{\mathbf{n}}_m &= \sin |\omega_{me}| = -\sin(\delta_{00}) \sin(\chi_m) + \\
 &\cos(\delta_{00}) \cos(\chi_m) \cos(\gamma_{00}) \left[\cos(\gamma_{00}) \sin |\omega_m| + \sin(\gamma_{00}) \frac{\omega_m}{|\omega_m|} \cos(\omega_m) \right] - \\
 &\cos(\delta_{00}) \cos(\chi_m) \sin(\gamma_{00}) \left[\cos(\gamma_{00}) \frac{\omega_m}{|\omega_m|} \cos(\omega_m) - \sin(\gamma_{00}) \sin |\omega_m| \right]
 \end{aligned} \tag{60}$$

or

$$\sin |\omega_{me}| = -\sin(\delta_{00}) \sin(\chi_m) + \cos(\delta_{00}) \cos(\chi_m) \sin |\omega_m| \tag{61}$$

gives the effective crystal angle, ω_{me} , in terms of the crystal setting angles, and of course

$$k_{me} = \frac{\pi}{d_M \sin |\omega_{me}|} \tag{62}$$

gives the monochromator element's wave-vector. Note that the vertical offset corrections for ω_{me} in terms of ω_m are second order, so that to first order

we can treat a vertically focussing crystal element by simply increasing the vertical mosaic. As long as χ_m is not too large we can take the signs of ω_{mi} and ω_{me} to be the same, so that

$$\begin{aligned}\widehat{\mathbf{o}}_m = & [\sin(\delta_{00}) + 2 \sin |\omega_{me}| \sin(\chi_m)] \widehat{\mathbf{z}} + \\ & \{\cos(\delta_{00}) \sin(\gamma_{00}) + 2 \sin(\omega_{me}) \cos(\chi_m) [\cos(\gamma_{00}) \cos(\omega_m) - \sin(\gamma_{00}) \sin(\omega_m)]\} \widehat{\mathbf{y}}_M + \\ & \{\cos(\delta_{00}) \cos(\gamma_{00}) - 2 \sin(\omega_{me}) \cos(\chi_m) [\cos(\gamma_{00}) \sin(\omega_m) + \sin(\gamma_{00}) \cos(\omega_m)]\} \widehat{\mathbf{x}}_M\end{aligned}\quad (63)$$

Recall that γ_{10} is supposed to be the deviation of the inplane part of this outgoing direction with respect to the monochromator to sample line, so we convert to those coordinates, $\widehat{\mathbf{x}}_S$ and $\widehat{\mathbf{y}}_S$. Compare this to

$$\mathbf{u}_{ms} = \sin(\delta_{10}) \widehat{\mathbf{z}} + \cos(\delta_{10}) [\sin(\gamma_{10}) \widehat{\mathbf{y}}_S + \cos(\gamma_{10}) \widehat{\mathbf{x}}_S] \quad (64)$$

so that

$$\sin(\delta_{10}) = \sin(\delta_{00}) + 2 \sin |\omega_{me}| \sin(\chi_m) \quad (65)$$

and

$$\begin{aligned}\cos(\delta_{10}) \sin(\gamma_{10}) = & \cos(2\theta_M) \{\cos(\delta_{00}) \sin(\gamma_{00}) + 2 \sin(\omega_{me}) \cos(\chi_m) [\cos(\gamma_{00}) \cos(\omega_m) - \sin(\gamma_{00}) \sin(\omega_m)]\} - \\ & \sin(2\theta_M) \{\cos(\delta_{00}) \cos(\gamma_{00}) - 2 \sin(\omega_{me}) \cos(\chi_m) [\cos(\gamma_{00}) \sin(\omega_m) + \sin(\gamma_{00}) \cos(\omega_m)]\}\end{aligned}\quad (66)$$

This replaces equation 11, and reduces to that equation when $\delta_{10} = \chi_m = 0$.

If the analyzer crystal element is displaced vertically so that its vertical deviation angle with respect to the nominal scattering plane is δ_{20} , and rotated about its vertical axis by the angle ω_{ai} with respect to the original line from sample to analyzer and then rotated by χ_a with respect to the crystal element's horizontal axis, we find that the crystal normal is pointing at

$$\widehat{\mathbf{n}}_a = \sin(\chi_a) \widehat{\mathbf{z}} + \cos(\chi_a) \left[\frac{\omega_a}{|\omega_a|} \cos(\omega_a) \widehat{\mathbf{y}}_a - \sin |\omega_a| \widehat{\mathbf{x}}_a \right] \quad (67)$$

where $\widehat{\mathbf{x}}_a$ and $\widehat{\mathbf{y}}_a$ are the unit vectors along the offset path and perpendicular to it so that

$$\widehat{\mathbf{x}}_a = \cos(\gamma_{20}) \widehat{\mathbf{x}}_A + \sin(\gamma_{20}) \widehat{\mathbf{y}}_A \quad (68)$$

$$\hat{\mathbf{y}}_a = -\sin(\gamma_{20})\hat{\mathbf{x}}_A + \cos(\gamma_{20})\hat{\mathbf{y}}_A \quad (69)$$

and so

$$\begin{aligned} \hat{\mathbf{n}}_a = \sin(\chi_a)\hat{\mathbf{z}} - \cos(\chi_a) \left[\cos(\gamma_{20})\sin|\omega_a| + \sin(\gamma_{20})\frac{\omega_a}{|\omega_a|}\cos(\omega_a) \right] \hat{\mathbf{x}}_A + \\ \cos(\chi_a) \left[\cos(\gamma_{20})\frac{\omega_a}{|\omega_a|}\cos(\omega_a) - \sin(\gamma_{20})\sin|\omega_a| \right] \hat{\mathbf{y}}_A \end{aligned} \quad (70)$$

The incident nominal direction upon this crystal element is given by,

$$\hat{\mathbf{i}}_a = \sin(\delta_{20})\hat{\mathbf{z}} + \cos(\delta_{20})[\sin(\gamma_{20})\hat{\mathbf{y}}_A + \cos(\gamma_{20})\hat{\mathbf{x}}_A] \quad (71)$$

so that

$$\hat{\mathbf{o}}_a = \hat{\mathbf{i}}_a + 2\sin|\omega_{ae}|\hat{\mathbf{n}}_a \quad (72)$$

and

$$\begin{aligned} \hat{\mathbf{o}}_a = [\sin(\delta_{20}) + 2\sin|\omega_{ae}|\sin(\chi_a)]\hat{\mathbf{z}} + \\ \left\{ \cos(\delta_{20})\sin(\gamma_{20}) + 2\sin|\omega_{ae}|\cos(\chi_a) \left[\cos(\gamma_{20})\frac{\omega_a}{|\omega_a|}\cos(\omega_a) - \sin(\gamma_{20})\sin|\omega_a| \right] \right\} \hat{\mathbf{y}}_A + \\ \left\{ \cos(\delta_{20})\cos(\gamma_{20}) - 2\sin|\omega_{ae}|\cos(\chi_a) \left[\cos(\gamma_{20})\sin|\omega_a| + \sin(\gamma_{20})\frac{\omega_a}{|\omega_a|}\cos(\omega_a) \right] \right\} \hat{\mathbf{x}}_A \end{aligned} \quad (73)$$

Referring to the diagram for the offset monochromator,

$$\begin{aligned} -\hat{\mathbf{i}}_a \bullet \hat{\mathbf{n}}_a = \sin|\omega_{ae}| = -\sin(\delta_{20})\sin(\chi_a) + \\ \cos(\delta_{20})\cos(\chi_a)\cos(\gamma_{20}) \left[\cos(\gamma_{20})\sin|\omega_a| + \sin(\gamma_{20})\frac{\omega_a}{|\omega_a|}\cos(\omega_a) \right] - \\ \cos(\delta_{20})\cos(\chi_a)\sin(\gamma_{20}) \left[\cos(\gamma_{20})\frac{\omega_a}{|\omega_a|}\cos(\omega_a) - \sin(\gamma_{20})\sin|\omega_a| \right] \end{aligned} \quad (74)$$

or

$$\sin|\omega_{ae}| = -\sin(\delta_{20})\sin(\chi_a) + \cos(\delta_{20})\cos(\chi_a)\sin|\omega_a| \quad (75)$$

gives the effective crystal angle, ω_{ae} , in terms of the crystal setting angles, and of course

$$k_{ae} = \frac{\pi}{d_A \sin|\omega_{ae}|} \quad (76)$$

gives the analyzer element's wave-vector. Note that the vertical offset corrections are second order, so that to first order we can treat a vertically focussing crystal element by simply increasing the vertical mosaic. As long as χ_a is not too large we can take the signs of ω_{ai} and ω_{ae} to be the same, so that

$$\begin{aligned} \hat{\mathbf{o}}_a = & [\sin(\delta_{20}) + 2 \sin |\omega_{ae}| \sin(\chi_a)] \hat{\mathbf{z}} + \\ & \{\cos(\delta_{20}) \sin(\gamma_{20}) + 2 \sin(\omega_{ae}) \cos(\chi_a) [\cos(\gamma_{20}) \cos(\omega_a) - \sin(\gamma_{20}) \sin(\omega_a)]\} \hat{\mathbf{y}}_A + \\ & \{\cos(\delta_{20}) \cos(\gamma_{20}) - 2 \sin(\omega_{ae}) \cos(\chi_a) [\cos(\gamma_{20}) \sin(\omega_a) + \sin(\gamma_{20}) \cos(\omega_a)]\} \hat{\mathbf{x}}_A \end{aligned} \quad (77)$$

Recall that γ_{30} is supposed to be the deviation of the inplane part of this outgoing direction with respect to the analyzer to detector line, so we convert to those coordinates, $\hat{\mathbf{x}}_D$ and $\hat{\mathbf{y}}_D$. Compare this

$$\mathbf{u}_{ad} = \sin(\delta_{30}) \hat{\mathbf{z}} + \cos(\delta_{30}) [\sin(\gamma_{30}) \hat{\mathbf{y}}_D + \cos(\gamma_{30}) \hat{\mathbf{x}}_D] \quad (78)$$

so that

$$\sin(\delta_{30}) = \sin(\delta_{20}) + 2 \sin |\omega_{ae}| \sin(\chi_a) \quad (79)$$

and

$$\begin{aligned} \cos(\delta_{30}) \sin(\gamma_{30}) = & \\ \cos(2\theta_A) \{ & \cos(\delta_{20}) \sin(\gamma_{20}) + 2 \sin(\omega_{ae}) \cos(\chi_a) [\cos(\gamma_{20}) \cos(\omega_a) - \sin(\gamma_{20}) \sin(\omega_a)] \} - \\ \sin(2\theta_A) \{ & \cos(\delta_{20}) \cos(\gamma_{20}) - 2 \sin(\omega_{ae}) \cos(\chi_a) [\cos(\gamma_{20}) \sin(\omega_a) + \sin(\gamma_{20}) \cos(\omega_a)] \} \end{aligned} \quad (80)$$

This replaces equation 11, and reduces to that equation when $\delta_{30} = \chi_a = 0$, and also gives the equations relating δ_{30} to δ_{20} and γ_{30} to γ_{20} ,

$$\sin(\delta_{30}) = \sin(\delta_{20}) + 2 \sin |\omega_{ae}| \sin(\chi_a) \quad (81)$$

Also, recall

$$-\hat{\mathbf{i}}_a \bullet \hat{\mathbf{n}}_a = \sin |\omega_{ae}| = -\sin(\delta_{20}) \sin(\chi_a) + \cos(\delta_{20}) \cos(\chi_a) \sin |\omega_a| \quad (82)$$

gives the effective crystal angle, ω_{ae} , in terms of the crystal setting angles, and of course

$$k_{ae} = \frac{\pi}{d_A \sin |\omega_{ae}|} \quad (83)$$

gives the analyzer element's wave-vector. In the end we have that the nominal path wave-vector impinging on the sample is

$$\mathbf{k}_{Ie} = k_{me} \mathbf{u}_{ms} \quad (84)$$

which requires inputs of δ_{00} , γ_{00} , χ_m and $\omega_m = \omega_{mi} - \gamma_{00}$, where recall that ω_{mi} is the crystal rotation angle with respect to the original reactor to monochromator line. Similarly, the nominal path wave-vector leaving the sample is

$$\mathbf{k}_{Fe} = k_{ae} \mathbf{u}_{sa} \quad (85)$$

which requires inputs of δ_{20} , γ_{20} , χ_a and $\omega_a = \omega_{ai} - \gamma_{20}$, where recall that ω_{ai} is the crystal rotation angle with respect to the original sample to analyzer line.

Now, as usual we would like to separate the horizontal and vertical transmission functions. Note however, that when crystals are displaced vertically and/or tilted, the crystal transmission function, equation 43, should use the horizontal deviation angles γ_{0m} and γ_{1m} not in the original scattering plane, but instead in the effective-path scattering plane of the crystal, and the vertical deviation angles should be out of this plane. Thus we need to complete the coordinate system for the effective-path scattering plane,

$$\hat{\mathbf{i}}_{m\perp} = [\hat{\mathbf{o}}_m - \cos(2\omega_{me}) \hat{\mathbf{i}}_m] / \sin(2\omega_{me}) \quad (86)$$

$$\hat{\mathbf{o}}_{m\perp} = [\cos(2\omega_{me}) \hat{\mathbf{o}}_m - \hat{\mathbf{i}}_m] / \sin(2\omega_{me}) \quad (87)$$

$$\hat{\mathbf{z}}_m = \hat{\mathbf{i}}_m \times \hat{\mathbf{o}}_m / \sin(2\omega_{me}) \quad (88)$$

and then we write the deviated pathes in the $\hat{\mathbf{i}}_m$ and $\hat{\mathbf{o}}_m$ scattering plane as

$$\hat{\mathbf{i}}_{md} = \sin(\delta_{0m}) \hat{\mathbf{z}}_m + \cos(\delta_{0m}) [\cos(\gamma_{0m}) \hat{\mathbf{i}}_m + \sin(\gamma_{0m}) \hat{\mathbf{i}}_{m\perp}] \quad (89)$$

$$\hat{\mathbf{o}}_{md} = \sin(\delta_{1m}) \hat{\mathbf{z}}_m + \cos(\delta_{1m}) [\cos(\gamma_{1m}) \hat{\mathbf{o}}_m + \sin(\gamma_{1m}) \hat{\mathbf{o}}_{m\perp}] \quad (90)$$

If we are to use these deviation angles in the TPF, we need to find γ_0 , γ_1 , δ_0 and δ_1 . The deviation pathes in terms of these angles are

$$\widehat{\mathbf{i}}_{md} = \sin(\delta_0) \widehat{\mathbf{z}} + \cos(\delta_0) [\cos(\gamma_0) \widehat{\mathbf{x}}_M + \sin(\gamma_0) \widehat{\mathbf{y}}_M] \quad (91)$$

$$\widehat{\mathbf{o}}_{md} = \sin(\delta_1) \widehat{\mathbf{z}} + \cos(\delta_1) [\cos(\gamma_1) \widehat{\mathbf{x}}_S + \sin(\gamma_1) \widehat{\mathbf{y}}_S] \quad (92)$$

Then

$$\sin(\delta_0) = \sin(\delta_{0m}) \widehat{\mathbf{z}} \bullet \widehat{\mathbf{z}}_m + \cos(\delta_{0m}) \left[\cos(\gamma_{0m}) \widehat{\mathbf{z}} \bullet \widehat{\mathbf{i}}_m + \sin(\gamma_{0m}) \widehat{\mathbf{z}} \bullet \widehat{\mathbf{i}}_{m\perp} \right] \quad (93)$$

$$\cos(\delta_0) \sin(\gamma_0) = \sin(\delta_{0m}) \widehat{\mathbf{y}}_M \bullet \widehat{\mathbf{z}}_m + \cos(\delta_{0m}) \left[\cos(\gamma_{0m}) \widehat{\mathbf{y}}_M \bullet \widehat{\mathbf{i}}_m + \sin(\gamma_{0m}) \widehat{\mathbf{y}}_M \bullet \widehat{\mathbf{i}}_{m\perp} \right] \quad (94)$$

$$\sin(\delta_1) = \sin(\delta_{1m}) \widehat{\mathbf{z}} \bullet \widehat{\mathbf{z}}_m + \cos(\delta_{1m}) [\cos(\gamma_{1m}) \widehat{\mathbf{z}} \bullet \widehat{\mathbf{o}}_m + \sin(\gamma_{1m}) \widehat{\mathbf{z}} \bullet \widehat{\mathbf{o}}_{m\perp}] \quad (95)$$

$$\cos(\delta_1) \sin(\gamma_1) = \sin(\delta_{1m}) \widehat{\mathbf{y}}_S \bullet \widehat{\mathbf{z}}_m + \cos(\delta_{1m}) [\cos(\gamma_{1m}) \widehat{\mathbf{y}}_S \bullet \widehat{\mathbf{o}}_m + \sin(\gamma_{1m}) \widehat{\mathbf{y}}_S \bullet \widehat{\mathbf{o}}_{m\perp}] \quad (96)$$

Unfortunately, we find that the original deviation angles are mixtures of both horizontal and vertical deviation angles with respect to the crystal scattering plane. However, if we keep only the first order corrections, then

$$\delta_0 = \delta_{0m} + \delta_{00} \quad (97)$$

$$\gamma_0 = \gamma_{0m} + \gamma_{00} \quad (98)$$

and

$$\delta_1 = \delta_{1m} + \delta_{00} + 2 \sin|\omega_{me}| \chi_m \quad (99)$$

$$\gamma_1 = \gamma_{1m} + \gamma_{00} + 2(\omega_m - \omega_M) \quad (100)$$

$$\gamma_{0m} = \gamma_{1m} + 2\overline{\Delta k_{me}} \tan(\omega_{me}) \quad (101)$$

where we have also used the fact that to first order $\omega_{me} = \omega_m$ when computing $\hat{\mathbf{y}}_S \bullet \hat{\mathbf{o}}_m$. Then equation 23 becomes

$$A_{Mh}^2 = \left(\frac{\gamma_0}{\alpha_0}\right)^2 + \left(\frac{\gamma_0 + \gamma_1 - 2\gamma_{00} - 2\Delta\omega_m}{2\eta_M}\right)^2 + \left(\frac{\gamma_1}{\alpha_1}\right)^2 + \left(\frac{\gamma_0 - \gamma_{00}}{\alpha_{RM}}\right)^2 + \left(\frac{\gamma_1 - \gamma_{00} - 2\Delta\omega_m}{\alpha_{MS}}\right)^2 \quad (102)$$

or using

$$\gamma_0 = \gamma_1 + 2\overline{\Delta k_m} \tan(\omega_m) - 2\Delta\omega_m$$

$$A_{Mh}^2 = \left(\frac{\gamma_1 + 2\overline{\Delta k_{me}} \tan(\omega_{me}) - 2\Delta\omega_m}{\alpha_0}\right)^2 + \left(\frac{\gamma_1 + \overline{\Delta k_{me}} \tan(\omega_{me}) - \gamma_{10}}{\eta_M}\right)^2 + \left(\frac{\gamma_1}{\alpha_1}\right)^2 + \left(\frac{\gamma_1 + 2\overline{\Delta k_{me}} \tan(\omega_{me}) - \gamma_{10}}{\alpha_{RM}}\right)^2 + \left(\frac{\gamma_1 - \gamma_{10}}{\alpha_{MS}}\right)^2 \quad (103)$$

where

$$\overline{\Delta k_{me}} = \frac{\Delta k_m}{k_{me}} \quad (104)$$

and

$$\Delta k_m = k_m - k_{me} \quad (105)$$

Similarly for the vertical part,

$$A_{Mv}^2 = \left(\frac{\delta_0}{\beta_0}\right)^2 + \left(\frac{\delta_{1m} - \delta_{0m}}{2\eta_{Mv} \sin(\omega_{me})}\right)^2 + \left(\frac{\delta_1}{\beta_1}\right)^2 + \left(\frac{\delta_{0m}}{\beta_{RM}}\right)^2 + \left(\frac{\delta_{1m}}{\beta_{MS}}\right)^2 \quad (106)$$

or

$$A_{Mv}^2 = \left(\frac{\delta_0}{\beta_0}\right)^2 + \left(\frac{\delta_1 - \delta_0 - 2\tilde{\chi}_m}{2\eta_{Mv} \sin(\omega_{me})}\right)^2 + \left(\frac{\delta_1}{\beta_1}\right)^2 + \left(\frac{\delta_0 - \delta_{00}}{\beta_{RM}}\right)^2 + \left(\frac{\delta_1 - \delta_{00} - 2\tilde{\chi}_m}{\beta_{MS}}\right)^2 \quad (107)$$

where we use

$$\tilde{\chi}_m = \sin |\omega_{me}| \chi_m \quad (108)$$

This exposition can be repeated exactly for the analyzer system, with the exception that the horizontal deviation angle after the analyzer is eliminated instead of the horizontal deviation angle before the monochromator so that

$$\hat{\mathbf{i}}_{a\perp} = \left[\hat{\mathbf{o}}_a - \cos(2\omega_{ae}) \hat{\mathbf{i}}_a \right] / \sin(2\omega_{ae}) \quad (109)$$

$$\hat{\mathbf{o}}_{a\perp} = \left[\cos(2\omega_{ae}) \hat{\mathbf{o}}_a - \hat{\mathbf{i}}_a \right] / \sin(2\omega_{ae}) \quad (110)$$

$$\hat{\mathbf{z}}_a = \hat{\mathbf{i}}_a \times \hat{\mathbf{o}}_a / \sin(2\omega_{ae}) \quad (111)$$

and then we write the deviated pathes in the $\hat{\mathbf{i}}_a$ and $\hat{\mathbf{o}}_a$ scattering plane as

$$\hat{\mathbf{i}}_{ad} = \sin(\delta_{2a}) \hat{\mathbf{z}}_a + \cos(\delta_{2a}) \left[\cos(\gamma_{2a}) \hat{\mathbf{i}}_a + \sin(\gamma_{2a}) \hat{\mathbf{i}}_{a\perp} \right] \quad (112)$$

$$\hat{\mathbf{o}}_{ad} = \sin(\delta_{3a}) \hat{\mathbf{z}}_a + \cos(\delta_{3a}) \left[\cos(\gamma_{3a}) \hat{\mathbf{o}}_a + \sin(\gamma_{3a}) \hat{\mathbf{o}}_{a\perp} \right] \quad (113)$$

If we are to use these deviation angles in the TPF, we need to find γ_2 , γ_3 , δ_2 and δ_3 . The deviation pathes in terms of these angles are

$$\hat{\mathbf{i}}_{ad} = \sin(\delta_2) \hat{\mathbf{z}} + \cos(\delta_2) \left[\cos(\gamma_2) \hat{\mathbf{x}}_A + \sin(\gamma_2) \hat{\mathbf{y}}_A \right] \quad (114)$$

$$\hat{\mathbf{o}}_{ad} = \sin(\delta_3) \hat{\mathbf{z}} + \cos(\delta_3) \left[\cos(\gamma_3) \hat{\mathbf{x}}_D + \sin(\gamma_3) \hat{\mathbf{y}}_D \right] \quad (115)$$

Then

$$\sin(\delta_2) = \sin(\delta_{2a}) \hat{\mathbf{z}} \bullet \hat{\mathbf{z}}_a + \cos(\delta_{2a}) \left[\cos(\gamma_{2a}) \hat{\mathbf{z}} \bullet \hat{\mathbf{i}}_a + \sin(\gamma_{2a}) \hat{\mathbf{z}} \bullet \hat{\mathbf{i}}_{a\perp} \right] \quad (116)$$

$$\cos(\delta_2) \sin(\gamma_2) = \sin(\delta_{2a}) \hat{\mathbf{y}}_A \bullet \hat{\mathbf{z}}_a + \cos(\delta_{2a}) \left[\cos(\gamma_{2a}) \hat{\mathbf{y}}_A \bullet \hat{\mathbf{i}}_a + \sin(\gamma_{2a}) \hat{\mathbf{y}}_A \bullet \hat{\mathbf{i}}_{a\perp} \right] \quad (117)$$

$$\sin(\delta_3) = \sin(\delta_{3a}) \hat{\mathbf{z}} \bullet \hat{\mathbf{z}}_a + \cos(\delta_{3a}) \left[\cos(\gamma_{3a}) \hat{\mathbf{z}} \bullet \hat{\mathbf{o}}_a + \sin(\gamma_{3a}) \hat{\mathbf{z}} \bullet \hat{\mathbf{o}}_{a\perp} \right] \quad (118)$$

$$\cos(\delta_3) \sin(\gamma_3) = \sin(\delta_{3a}) \hat{\mathbf{y}}_D \bullet \hat{\mathbf{z}}_a + \cos(\delta_{3a}) [\cos(\gamma_{3a}) \hat{\mathbf{y}}_D \bullet \hat{\mathbf{o}}_a + \sin(\gamma_{3a}) \hat{\mathbf{y}}_D \bullet \hat{\mathbf{o}}_{a\perp}] \quad (119)$$

Unfortunately, we find that the original deviation angles are mixtures of both horizontal and vertical deviation angles with respect to the crystal scattering plane. However, if we keep only the first order corrections, then

$$\delta_2 = \delta_{2a} + \delta_{20} \quad (120)$$

$$\gamma_2 = \gamma_{2a} + \gamma_{20} \quad (121)$$

and

$$\delta_3 = \delta_{3a} + \delta_{20} + 2 \sin |\omega_{ae}| \chi_a \quad (122)$$

$$\gamma_3 = \gamma_{3a} + \gamma_{20} + 2(\omega_a - \omega_A) \quad (123)$$

$$\gamma_{3a} = \gamma_{2a} - 2\overline{\Delta k_{ae}} \tan(\omega_{ae}) \quad (124)$$

where we have also used the fact that to first order $\omega_{me} = \omega_m$ when computing $\hat{\mathbf{y}}_S \bullet \hat{\mathbf{o}}_m$.

$$A_{Ah}^2 = \left(\frac{\gamma_2}{\alpha_2}\right)^2 + \left(\frac{\gamma_2 + \gamma_3 - 2\gamma_{20} - 2\Delta\omega_a}{2\eta_A}\right)^2 + \left(\frac{\gamma_3}{\alpha_3}\right)^2 + \left(\frac{\gamma_2 - \gamma_{20}}{\alpha_{SA}}\right)^2 + \left(\frac{\gamma_3 - \gamma_{20} - 2\Delta\omega_a}{\alpha_{AD}}\right)^2 \quad (125)$$

or using

$$\gamma_3 = \gamma_2 - 2\overline{\Delta k_{ae}} \tan(\omega_{ae}) + 2\Delta\omega_a \quad (126)$$

$$A_{Ah}^2 = \left(\frac{\gamma_2}{\alpha_2}\right)^2 + \left(\frac{\gamma_2 - \overline{\Delta k_{ae}} \tan(\omega_{ae}) - \gamma_{20}}{\eta_A}\right)^2 + \left(\frac{\gamma_2 - 2\overline{\Delta k_{ae}} \tan(\omega_{ae}) + 2\Delta\omega_a}{\alpha_3}\right)^2 + \left(\frac{\gamma_2 - \gamma_{20}}{\alpha_{SA}}\right)^2 + \left(\frac{\gamma_2 - 2\overline{\Delta k_{ae}} \tan(\omega_{ae}) - \gamma_{20}}{\alpha_{AD}}\right)^2 \quad (127)$$

and for the vertical analyzer part

$$A_{Av}^2 = \left(\frac{\delta_2}{\beta_2}\right)^2 + \left(\frac{\delta_{3a} - \delta_{2a}}{2\eta_{Av} \sin(\omega_{ae})}\right)^2 + \left(\frac{\delta_3}{\beta_3}\right)^2 + \left(\frac{\delta_{2a}}{\beta_{SA}}\right)^2 + \left(\frac{\delta_{3a}}{\beta_{AD}}\right)^2 \quad (128)$$

or

$$A_{Av}^2 = \left(\frac{\delta_2}{\beta_2}\right)^2 + \left(\frac{\delta_3 - \delta_2 - 2\tilde{\chi}_a}{2\eta_{Av} \sin(\omega_{ae})}\right)^2 + \left(\frac{\delta_3}{\beta_3}\right)^2 + \left(\frac{\delta_2 - \delta_{20}}{\beta_{SA}}\right)^2 + \left(\frac{\delta_3 - \delta_{20} - 2\tilde{\chi}_a}{\beta_{AD}}\right)^2 \quad (129)$$

where

$$\tilde{\chi}_a = \sin|\omega_{ae}| \chi_a \quad (130)$$

For compatibility, we have

$$\vartheta_M = \omega_M - \omega_{mi} = -(\gamma_{00} + \Delta\omega_m) \quad (131)$$

$$\gamma_{10} = \gamma_{00} + 2\Delta\omega_m = -(\gamma_{00} + 2\vartheta_M) \quad (132)$$

$$\vartheta_A = \omega_A - \omega_{ai} = -(\gamma_{20} + \Delta\omega_a) \quad (133)$$

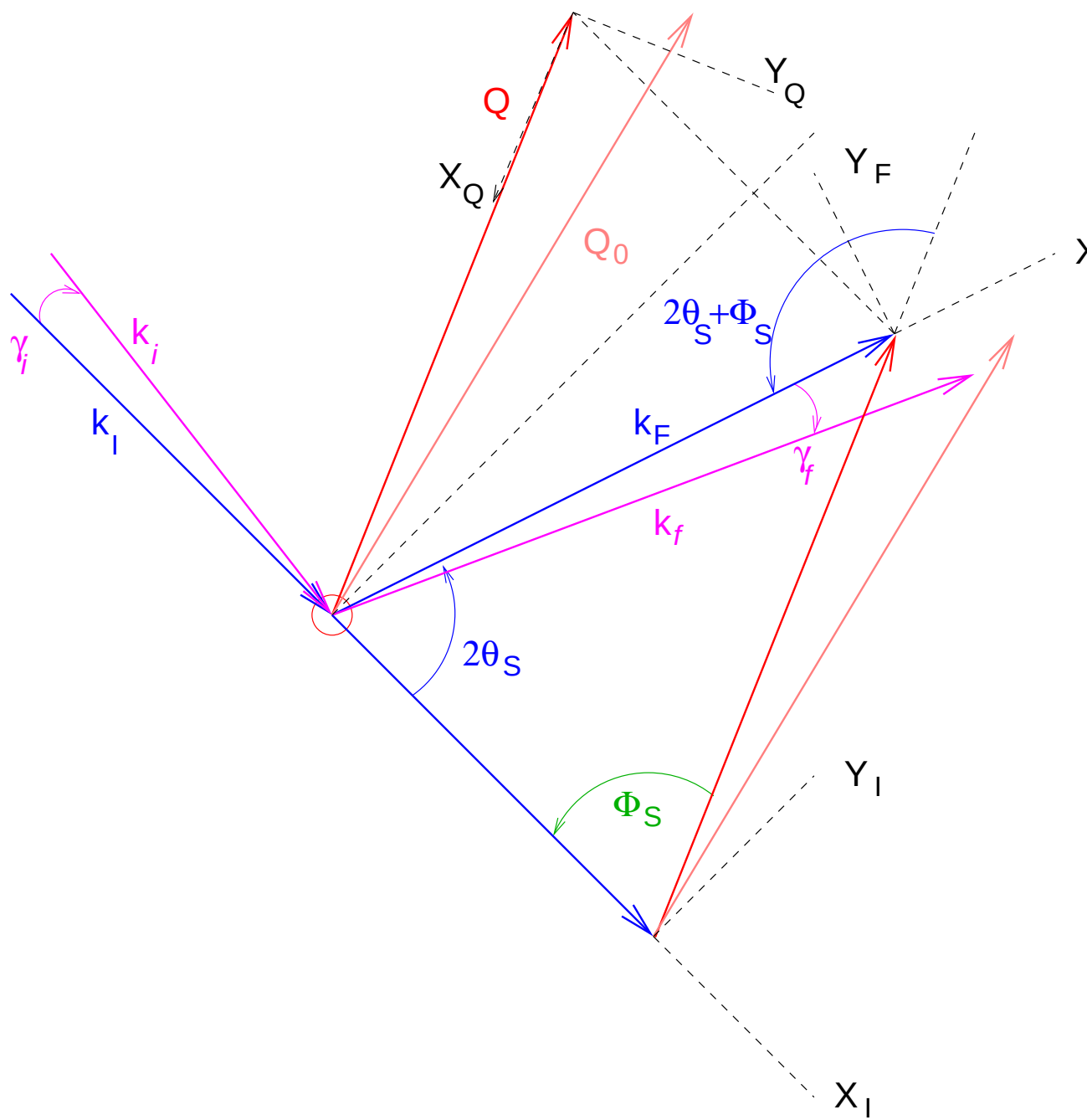
$$\gamma_{30} = \gamma_{20} + 2\Delta\omega_a = -(\gamma_{20} + 2\vartheta_A) \quad (134)$$

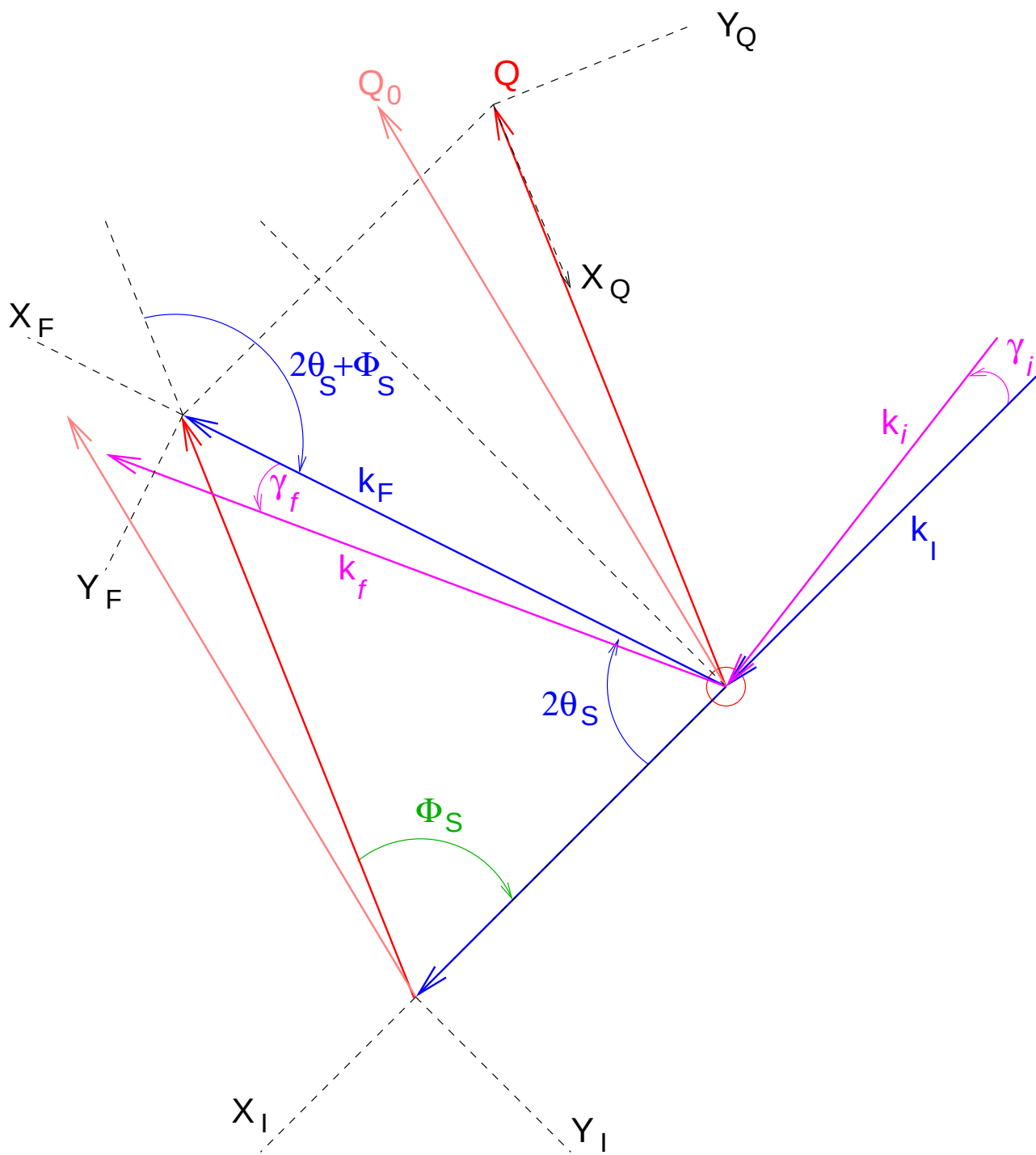
$$\delta_{10} = \delta_{00} + 2 \sin|\omega_{me}| \chi_m \quad (135)$$

$$\delta_{30} = \delta_{20} + 2 \sin|\omega_{ae}| \chi_a \quad (136)$$

horizontal resolution function for a triple-axis spectrometer

Now that we have shown that we can write the TPF for a spectrometer with displaced, rotated and tilted crystals in terms of the deviation angles with respect to the originally defined path, we can find the resolution function, which is the TPF function with variables changed to wave-vector and energy transfer of the neutron, and integration of the remaining degrees of freedom.





Above we show the two scattering diagrams for left and right scattering.

Now

$$\Delta \mathbf{k}_i = \mathbf{k}_i - \mathbf{k}_I = x_i \hat{\mathbf{x}}_I + y_i \hat{\mathbf{y}}_I + z_i \hat{\mathbf{z}} \quad (137)$$

$$\Delta \mathbf{k}_f = \mathbf{k}_f - \mathbf{k}_F = x_f \hat{\mathbf{x}}_F + y_f \hat{\mathbf{y}}_F + z_f \hat{\mathbf{z}} \quad (138)$$

$$x_i = \Delta k_i \quad (139)$$

$$y_i = \gamma_i k_{Im} = \gamma_i \frac{k_I}{\kappa_m} \quad (140)$$

$$z_i = \delta_i k_{Im} = \delta_i \frac{k_I}{\kappa_m} \quad (141)$$

with similar equations for x_f , y_f and z_f .

Note that

$$\Delta k_i = k_i - k_I = k_i - k_{me} + k_{me} - k_I = \Delta k_m + k_{me} - k_I$$

$$k_{me} = \frac{\pi}{d_M \sin |\omega_{me}|}$$

$$k_I = \frac{\pi}{d_M \sin |\omega_M|}$$

$$\sin |\omega_{me}| = -\sin(\delta_{00}) \sin(\chi_m) + \cos(\delta_{00}) \cos(\chi_m) \sin |\omega_m| \quad (142)$$

Note that to first order $\omega_{me} = \omega_m$.

$$\omega_m = \omega_{mi} - \gamma_{00}$$

$$\Delta \omega_m = \omega_m - \omega_M \quad (143)$$

$$x_i = \Delta k_m + k_{me} - k_I = \Delta k_m + \Delta k_I \quad (144)$$

$$\overline{\Delta k_{me}} = \frac{\Delta k_m}{k_{me}} = \frac{x_i - \Delta k_I}{k_{me}} = \frac{k_I}{k_{me}} \frac{x_i - \Delta k_I}{k_I} \quad (145)$$

$$x_f = \Delta k_a + k_{ae} - k_F = \Delta k_a + \Delta k_F \quad (146)$$

$$\overline{\Delta k_{ae}} = \frac{\Delta k_a}{k_{ae}} = \frac{x_f - \Delta k_F}{k_{ae}} = \frac{k_F}{k_{ae}} \frac{x_f - \Delta k_F}{k_F} \quad (147)$$

We can transform to the coordinate system which has $\mathbf{x}$ antiparallel to $\mathbf{Q}$ using,

$$\hat{\mathbf{x}}_I = \cos(\Phi_S) \hat{\mathbf{x}}_Q + \sin(\Phi_S) \hat{\mathbf{y}}_Q \quad (148)$$

$$\hat{\mathbf{y}}_I = -\sin(\Phi_S) \hat{\mathbf{x}}_Q + \cos(\Phi_S) \hat{\mathbf{y}}_Q \quad (149)$$

$$\hat{\mathbf{x}}_F = \cos(2\theta_S + \Phi_S) \hat{\mathbf{x}}_Q + \sin(2\theta_S + \Phi_S) \hat{\mathbf{y}}_Q \quad (150)$$

$$\hat{\mathbf{y}}_F = -\sin(2\theta_S + \Phi_S) \hat{\mathbf{x}}_Q + \cos(2\theta_S + \Phi_S) \hat{\mathbf{y}}_Q \quad (151)$$

Note that Φ_S and $2\theta_S$ have the same sign so that these transformations work for either left or right scattering. As in [CandN] define

$$a = \sin(\Phi_S) \quad (152)$$

$$b = \cos(\Phi_S) \quad (153)$$

$$A = \sin(2\theta_S + \Phi_S) \quad (154)$$

$$B = \cos(2\theta_S + \Phi_S) \quad (155)$$

so that

$$\Delta \mathbf{k}_i = (bx_i - ay_i) \hat{\mathbf{x}}_Q + (ax_i + by_i) \hat{\mathbf{y}}_Q + z_i \hat{\mathbf{z}} \quad (156)$$

$$\Delta \mathbf{k}_f = (Bx_f - Ay_f) \hat{\mathbf{x}}_Q + (Ax_f + By_f) \hat{\mathbf{y}}_Q + z_f \hat{\mathbf{z}} \quad (157)$$

Note that the terms with a and A are oppositely signed to those in [CandN] since we are using signed angles, and for right scattering as used in the exposition of [CandN], those angles are negative. Now writing the deviation of the wave-vector transferred to the neutron,

$$\Delta \mathbf{Q} = \Delta \mathbf{k}_f - \Delta \mathbf{k}_i \quad (158)$$

we have that

$$\Delta Q_x = Bx_f - Ay_f - bx_i + ay_i \quad (159)$$

$$\Delta Q_y = Ax_f + By_f - ax_i - by_i \quad (160)$$

$$\Delta Q_z = z_f - z_i \quad (161)$$

Also the nominal energy transferred to the sample is

$$E = D_n (k_I^2 - k_F^2) \quad (162)$$

where

$$D_n = \frac{\hbar^2}{2m_n} = 2.072141789 \text{ meV } \text{\AA}^2 \quad (163)$$

so that

$$\Delta E = 2D_n (x_i k_I - x_f k_F) \quad (164)$$

or

$$x_f = \lambda x_i - \frac{\Delta E}{2D_n k_F} \quad (165)$$

where $\lambda = k_I/k_F$. Now solving equations 157 and 158 for y_i and y_f

$$\Delta Q_x = B \left(\lambda x_i - \frac{\Delta E}{2D_n k_F} \right) - Ay_f - bx_i + ay_i \quad (166)$$

$$\Delta Q_y = A \left(\lambda x_i - \frac{\Delta E}{2D_n k_F} \right) + By_f - ax_i - by_i \quad (167)$$

or

$$C_x = \Delta Q_x + \frac{B\Delta E}{2D_n k_F} + (b - B\lambda) x_i = -Ay_f + ay_i \quad (168)$$

$$C_y = \Delta Q_y + \frac{A\Delta E}{2D_n k_F} + (a - A\lambda) x_i = +By_f - by_i \quad (169)$$

The determinant of the y_i and y_f coefficients is $Ab - aB = \sin(2\theta_S) = \alpha$. Also $bB + aA = \cos(2\theta_S) = \beta$ and $1/(2D_n k_F) = r$ so that

$$\alpha y_f = -C_x b - C_y a = -b\Delta Q_x - a\Delta Q_y - r\beta\Delta E - x_i(1 - \lambda\beta) \quad (170)$$

$$\alpha y_i = -C_x B - C_y A = -B\Delta Q_x - A\Delta Q_y - r\Delta E - x_i(\beta - \lambda) \quad (171)$$

As in [CandN] we set

$$y_i = Cx_i + D \quad (172)$$

$$y_f = Ex_i + F \quad (173)$$

$$x_f = \lambda x_i + H \quad (174)$$

and

$$\begin{pmatrix} D \\ F \\ H \end{pmatrix} = \begin{pmatrix} d_x & d_y & d_e \\ f_x & f_y & f_e \\ h_x & h_y & h_e \end{pmatrix} \begin{pmatrix} \Delta Q_x \\ \Delta Q_y \\ \Delta E \end{pmatrix} = \overline{\alpha} \bullet \Delta \quad (175)$$

where

$$\begin{pmatrix} d_x & d_y & d_e \\ f_x & f_y & f_e \\ h_x & h_y & h_e \end{pmatrix} = \overline{\alpha} = \begin{pmatrix} -B/\alpha & -A/\alpha & -r/\alpha \\ -b/\alpha & -a/\alpha & -r\beta/\alpha \\ 0 & 0 & -r \end{pmatrix} \quad (176)$$

and

$$C = (\lambda - \beta) / \alpha \quad (177)$$

$$E = (\lambda\beta - 1) / \alpha \quad (178)$$

Now we substitute into the transmission functions using

$$\overline{\Delta k_m} = \frac{\Delta k_m}{k_{me}} = \frac{x_i - \Delta k_I}{k_{me}} = \frac{k_I}{k_{me}} \frac{x_i - \Delta k_I}{k_I} = \kappa_m \frac{x_i - \Delta k_I}{k_I} \quad (179)$$

$$\overline{\Delta k_a} = \frac{\Delta k_a}{k_{ae}} = \frac{x_f - \Delta k_F}{k_{ae}} = \frac{k_F}{k_{ae}} \frac{x_f - \Delta k_F}{k_F} = \kappa_a \frac{x_f - \Delta k_F}{k_F} \quad (180)$$

$$A_{Mh}^2 = \left(\frac{\gamma_1 + 2\overline{\Delta k_{me}} \tan(\omega_{me}) - 2\Delta\omega_m}{\alpha_0} \right)^2 + \left(\frac{\gamma_1 + \overline{\Delta k_{me}} \tan(\omega_{me}) - \gamma_{10}}{\eta_M} \right)^2 + \left(\frac{\gamma_1}{\alpha_1} \right)^2 \\ + \left(\frac{\gamma_1 + 2\overline{\Delta k_{me}} \tan(\omega_{me}) - \gamma_{10}}{\alpha_{RM}} \right)^2 + \left(\frac{\gamma_1 - \gamma_{10}}{\alpha_{MS}} \right)^2 \quad (181)$$

$$A_{Mh}^2 = \left(\frac{\kappa_m y_i + 2\kappa_m (x_i - \Delta k_I) \tan(\omega_{me}) - 2\Delta\omega_m k_I}{\alpha_0 k_I} \right)^2 + \left(\frac{\kappa_m y_i + \kappa_m (x_i - \Delta k_I) \tan(\omega_{me}) - \gamma_{10} k_I}{\eta_M k_I} \right)^2 + \left(\frac{\kappa_m y_i}{\alpha_1 k_I} \right)^2 \\ + \left(\frac{\kappa_m y_i + 2\kappa_m (x_i - \Delta k_I) \tan(\omega_{me}) - \gamma_{10} k_I}{\alpha_{RM} k_I} \right)^2 + \left(\frac{\kappa_m y_i - \gamma_{10} k_I}{\alpha_{MS} k_I} \right)^2 \quad (182)$$

$$A_{Ah}^2 = \left(\frac{\gamma_2}{\alpha_2} \right)^2 + \left(\frac{\gamma_2 - \overline{\Delta k_{ae}} \tan(\omega_{ae}) - \gamma_{20}}{\eta_A} \right)^2 + \left(\frac{\gamma_2 - 2\overline{\Delta k_{ae}} \tan(\omega_{ae}) + 2\Delta\omega_a}{\alpha_3} \right)^2 \\ + \left(\frac{\gamma_2 - \gamma_{20}}{\alpha_{SA}} \right)^2 + \left(\frac{\gamma_2 - 2\overline{\Delta k_{ae}} \tan(\omega_{ae}) - \gamma_{20}}{\alpha_{AD}} \right)^2 \quad (183)$$

$$A_{Ah}^2 = \left(\frac{\kappa_a y_f - 2\kappa_a (x_f - \Delta k_F) \tan(\omega_{ae}) + 2\Delta\omega_a k_F}{\alpha_3 k_F} \right)^2 + \left(\frac{\kappa_a y_f - \kappa_a (x_f - \Delta k_F) \tan(\omega_{ae}) - \gamma_{20} k_F}{\eta_A k_F} \right)^2 + \left(\frac{\kappa_a y_f}{\alpha_2 k_F} \right)^2 \\ + \left(\frac{\kappa_a y_f - 2\kappa_a (x_f - \Delta k_F) \tan(\omega_{ae}) - \gamma_{20} k_F}{\alpha_{AD} k_F} \right)^2 + \left(\frac{\kappa_a y_f - \gamma_{20} k_F}{\alpha_{SA} k_F} \right)^2 \quad (184)$$

Note that the crystal mosaic parts of the TPF can be written

$$A_{Mh}^2 = \left(\frac{\mu_M x_i + \kappa_m D + \varepsilon_M k_I}{\eta_M k_I} \right)^2 \quad (185)$$

$$A_{Ah}^2 = \left(\frac{\mu_A x_i + \kappa_a F - \kappa_a H \tan(\omega_a) + \varepsilon_A k_F}{\eta_A k_F} \right)^2 \quad (186)$$

where

$$\mu_M = \kappa_m [C + \tan(\omega_m)] \quad (187)$$

$$\varepsilon_M = -\kappa_m \frac{\Delta k_I}{k_I} \tan(\omega_m) - \gamma_{10} = \kappa_m (\omega_m - \omega_M) - \gamma_{10} \quad (188)$$

$$\mu_A = \kappa_a [E - \lambda \tan(\omega_a)] \quad (189)$$

$$\varepsilon_A = \kappa_a \frac{\Delta k_F}{k_F} \tan(\omega_a) - \gamma_{20} = -\kappa_a (\omega_a - \omega_A) - \gamma_{20} \quad (190)$$

Then, when $\mu_M = 0$, the part of the TPF due to the monochromator becomes a constant over the infinite plane defined by

$$d_x \Delta Q_x + d_y \Delta Q_y + d_e \Delta E + \varepsilon_M k_I = 0 \quad (191)$$

and similarly for the analyzer, when $\mu_A = 0$, that part of the TPF becomes a constant over the infinite plane defined by

$$f_x \Delta Q_x + f_y \Delta Q_y + (f_e - \kappa_a h_e \tan(\omega_a)) \Delta E + \varepsilon_A k_F = 0 \quad (192)$$

This artifact is produced by a model based on angles which allows crystals of infinite spatial extent, and it can be removed by including the effects of geometrical collimation provided by finite sized crystals.

We can also estimate the offsets in $\mathbf{Q}$ and E using the expressions

$$\mathbf{Q} = \mathbf{k}_{Fe} - \mathbf{k}_{Ie}$$

$$E = D_n (k_{Ie}^2 - k_{Fe}^2)$$

$$\mathbf{k}_{Fe} - \mathbf{k}_{Ie} = \mathbf{Q}_o + \delta Q_z \hat{\mathbf{z}} + k_{Fe} \gamma_{20} \mathbf{u}_1 - k_{Ie} \gamma_{10} \mathbf{u}_2 + \Delta k_F \mathbf{v}_1 - \Delta k_I \mathbf{v}_2$$

where

$$\delta Q_z = k_{Fe}\delta_{20} - k_{Ie}\delta_{10}$$

$$\mathbf{u}_1 = -A\hat{\mathbf{x}}_Q + B\hat{\mathbf{y}}_Q$$

$$\mathbf{u}_2 = -a\hat{\mathbf{x}}_Q + b\hat{\mathbf{y}}_Q$$

$$\mathbf{v}_1 = B\hat{\mathbf{x}}_Q + A\hat{\mathbf{y}}_Q$$

$$\mathbf{v}_2 = b\mathbf{x}_Q + a\hat{\mathbf{y}}_Q$$

or

$$\delta Q_x = -Ak_{Fe}\gamma_{20} + ak_{Ie}\gamma_{10} + B\Delta k_F - b\Delta k_I$$

$$\delta Q_y = Bk_{Fe}\gamma_{20} - bk_{Ie}\gamma_{10} + A\Delta k_F - a\Delta k_I$$

Also

$$E/D_n = E_o/D_n + 2(k_I\Delta k_I - k_F\Delta k_F) + \Delta k_I^2 - \Delta k_F^2$$

$$E = E_o + D_n [\Delta k_I (k_{Ie} + k_I) - \Delta k_F (k_{Fe} + k_F)]$$

Now

$$A_{Mh}^2 = \left(a_1x_i + a_2y_i + \hat{\theta}_m\right)^2 + (a_3y_i)^2 + (\tilde{a}_3y_i - \hat{\gamma}_{10})^2 + \left(a_7x_i + a_8y_i + \hat{\theta}_0\right)^2 + (\tilde{a}_7x_i + \tilde{a}_8y_i - \hat{\gamma}_{00})^2 \quad (193)$$

$$A_{Ah}^2 = \left(a_5x_f + a_6y_f + \hat{\theta}_a\right)^2 + (a_4y_f)^2 + (\tilde{a}_4y_f - \hat{\gamma}_{20})^2 + \left(a_9x_f + a_{10}y_f + \hat{\theta}_3\right)^2 + (\tilde{a}_9x_f + \tilde{a}_{10}y_f - \hat{\gamma}_{30})^2 \quad (194)$$

The a_n coefficients, following [CandN] closely, are

$$a_1 = \kappa_m \tan(\omega_{me}) / \eta_M k_I \quad (195)$$

$$a_2 = \kappa_m / \eta_M k_I \quad (196)$$

$$a_3 = \kappa_m / \alpha_1 k_I \quad (197)$$

$$\tilde{a}_3 = \kappa_m / \alpha_{MS} k_I \quad (198)$$

$$a_4 = \kappa_a / \alpha_2 k_F \quad (199)$$

$$\tilde{a}_4 = \kappa_a / \alpha_{SA} k_F \quad (200)$$

$$a_5 = -\kappa_a \tan(\omega_{ae}) / \eta_A k_F \quad (201)$$

$$a_6 = \kappa_a / \eta_A k_F \quad (202)$$

$$a_7 = 2\kappa_m \tan(\omega_{me}) / \alpha_0 k_I \quad (203)$$

$$\tilde{a}_7 = 2\kappa_m \tan(\omega_{me}) / \alpha_{RM} k_I \quad (204)$$

$$a_8 = \kappa_m / \alpha_0 k_I \quad (205)$$

$$\tilde{a}_8 = \kappa_m / \alpha_{RM} k_I \quad (206)$$

$$a_9 = -2\kappa_a \tan(\omega_{ae}) / \alpha_3 k_F \quad (207)$$

$$\tilde{a}_9 = -2\kappa_a \tan(\omega_{ae}) / \alpha_{AD} k_F \quad (208)$$

$$a_{10} = \kappa_a / \alpha_3 k_F \quad (209)$$

$$\tilde{a}_{10} = \kappa_a / \alpha_{AD} k_F \quad (210)$$

We point out that a_5 , a_6 , a_9 and a_{10} have changed sign as compared to [CandN], because we are using the crystal rotation angles to handle the various scattering directions of the 3 crystals. Looking at the definition of the b_n coefficients below, it is found that these sign changes cancel. Also we have the correction parameters,

$$\hat{\theta}_m = -(\gamma_{10} - \Delta\omega_m + (1 - \kappa_m) \Delta\omega_m) / \eta_M = -(\gamma_{10} - \kappa_m \Delta\omega_m) / \eta_M = (\vartheta_M + \overline{\Delta\omega_m}) / \eta_M \quad (211)$$

$$\kappa_m \Delta\omega_m = (1 + \cot(\omega_M) \Delta\omega_m) \Delta\omega_m$$

$$\overline{\Delta\omega_m} = (\kappa_m - 1) \Delta\omega_m$$

$$\hat{\theta}_0 = 2\overline{\Delta\omega_m} / \alpha_0$$

$$\hat{\gamma}_{00} = (\gamma_{00} - 2\overline{\Delta\omega_m}) / \alpha_{RM} \quad (212)$$

$$\hat{\gamma}_{10} = (\gamma_{00} + 2\Delta\omega_m) / \alpha_{MS} = \gamma_{10} / \alpha_{MS} \quad (213)$$

$$\hat{\theta}_a = -(\gamma_{20} + \Delta\omega_a - (1 - \kappa_a) \Delta\omega_a) / \eta_A = -(\gamma_{20} + \kappa_a \Delta\omega_a) / \eta_A = (\vartheta_A - \overline{\Delta\omega_a}) / \eta_A \quad (214)$$

$$\kappa_a \Delta\omega_a = (1 + \cot(\omega_A) \Delta\omega_a) \Delta\omega_a$$

$$\overline{\Delta\omega_a} = (\kappa_a - 1) \Delta\omega_a$$

$$\hat{\theta}_3 = -2\overline{\Delta\omega_a} / \alpha_3$$

$$\hat{\gamma}_{20} = \gamma_{20} / \alpha_{SA} \quad (215)$$

$$\widehat{\gamma}_{30} = (\gamma_{30} + 2\overline{\Delta\omega_a}) / \alpha_{AD} \quad (216)$$

Now we can rewrite A_{Mh}^2 as a quadratic form in x_i and y_i ,

$$A_{Mh}^2 = \begin{pmatrix} x_i & y_i \end{pmatrix} \mathbf{P} \begin{pmatrix} x_i \\ y_i \end{pmatrix} + 2\mathbf{K} \bullet \begin{pmatrix} x_i \\ y_i \end{pmatrix} + C_{Mh} \quad (217)$$

where, using the b_n parameter definitions of [CandN],

$$\begin{aligned} \mathbf{P} &= \begin{pmatrix} a_1^2 + a_7^2 + \widetilde{a}_7^2 & a_1a_2 + a_7a_8 + \widetilde{a}_7\widetilde{a}_8 \\ a_1a_2 + a_7a_8 + \widetilde{a}_7\widetilde{a}_8 & a_2^2 + a_8^2 + a_3^2 + \widetilde{a}_8^2 + \widetilde{a}_3^2 \end{pmatrix} \\ &= \begin{pmatrix} b_5 + \widetilde{a}_7^2 & b_0 + \widetilde{a}_7\widetilde{a}_8 \\ b_0 + \widetilde{a}_7\widetilde{a}_8 & b_1 + \widetilde{a}_8^2 + \widetilde{a}_3^2 \end{pmatrix} = \begin{pmatrix} \widetilde{b}_5 & \widetilde{b}_0 \\ \widetilde{b}_0 & \widetilde{b}_1 \end{pmatrix} \end{aligned} \quad (218)$$

$$\mathbf{K}^t = \begin{pmatrix} a_1\widehat{\theta}_m + a_7\widehat{\theta}_0 - \widetilde{a}_7\widehat{\gamma}_{00} & a_2\widehat{\theta}_m + a_8\widehat{\theta}_0 - \widetilde{a}_8\widehat{\gamma}_{00} - \widetilde{a}_3\widehat{\gamma}_{10} \end{pmatrix} \quad (219)$$

$$C_{Mh} = \widehat{\theta}_m^2 + \widehat{\theta}_0^2 + \widehat{\gamma}_{00}^2 + \widehat{\gamma}_{10}^2 \quad (220)$$

Similarly, for the analyzer horizontal part

$$A_{Ah}^2 = \begin{pmatrix} x_f & y_f \end{pmatrix} \mathbf{Q} \begin{pmatrix} x_f \\ y_f \end{pmatrix} + 2\mathbf{L} \bullet \begin{pmatrix} x_f \\ y_f \end{pmatrix} + C_{Ah} \quad (221)$$

where

$$\begin{aligned} \mathbf{Q} &= \begin{pmatrix} a_5^2 + a_9^2 + \widetilde{a}_9^2 & a_5a_6 + a_9a_{10} + \widetilde{a}_9\widetilde{a}_{10} \\ a_5a_6 + a_9a_{10} + \widetilde{a}_9\widetilde{a}_{10} & a_6^2 + a_4^2 + a_{10}^2 + \widetilde{a}_4^2 + \widetilde{a}_{10}^2 \end{pmatrix} \\ &= \begin{pmatrix} b_3 + \widetilde{a}_9^2 & b_4 + \widetilde{a}_9\widetilde{a}_{10} \\ b_4 + \widetilde{a}_9\widetilde{a}_{10} & b_2 + \widetilde{a}_4^2 + \widetilde{a}_{10}^2 \end{pmatrix} = \begin{pmatrix} \widetilde{b}_3 & \widetilde{b}_4 \\ \widetilde{b}_4 & \widetilde{b}_2 \end{pmatrix} \end{aligned} \quad (222)$$

$$\mathbf{L}^t = \begin{pmatrix} a_5\widehat{\theta}_a + a_9\widehat{\theta}_3 - \widetilde{a}_9\widehat{\gamma}_{30} & a_6\widehat{\theta}_a + a_{10}\widehat{\theta}_3 - \widetilde{a}_{10}\widehat{\gamma}_{30} - \widetilde{a}_4\widehat{\gamma}_{20} \end{pmatrix} \quad (223)$$

$$C_{Ah} = \widehat{\theta}_a^2 + \widehat{\theta}_3^2 + \widehat{\gamma}_{30}^2 + \widehat{\gamma}_{20}^2 \quad (224)$$

Now use the relations 170, 171 and 172 to do the substitutions,

$$\begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ C & D \end{pmatrix} \begin{pmatrix} x_i \\ 1 \end{pmatrix} \quad (225)$$

$$\begin{pmatrix} x_f \\ y_f \end{pmatrix} = \begin{pmatrix} \lambda & H \\ E & F \end{pmatrix} \begin{pmatrix} x_i \\ 1 \end{pmatrix} \quad (226)$$

so that the quadratic forms become

$$A_{Mh}^2 = \begin{pmatrix} x_i & 1 \end{pmatrix} \begin{pmatrix} 1 & C \\ 0 & D \end{pmatrix} \mathbf{P} \begin{pmatrix} 1 & 0 \\ C & D \end{pmatrix} \begin{pmatrix} x_i \\ 1 \end{pmatrix} + 2\mathbf{K} \bullet \begin{pmatrix} 1 & 0 \\ C & D \end{pmatrix} \begin{pmatrix} x_i \\ 1 \end{pmatrix} + C_{Mh} \quad (227)$$

$$A_{Ah}^2 = \begin{pmatrix} x_i & 1 \end{pmatrix} \begin{pmatrix} \lambda & E \\ H & F \end{pmatrix} \mathbf{Q} \begin{pmatrix} \lambda & H \\ E & F \end{pmatrix} \begin{pmatrix} x_i \\ 1 \end{pmatrix} + 2\mathbf{L} \bullet \begin{pmatrix} \lambda & H \\ E & F \end{pmatrix} \begin{pmatrix} x_i \\ 1 \end{pmatrix} + C_{Ah} \quad (228)$$

which we rewrite as

$$A_{Mh}^2 = \begin{pmatrix} x_i & 1 \end{pmatrix} \mathbf{S} \begin{pmatrix} x_i \\ 1 \end{pmatrix} + 2\mathbf{U}^t \begin{pmatrix} x_i \\ 1 \end{pmatrix} + C_{Mh} \quad (229)$$

$$A_{Ah}^2 = \begin{pmatrix} x_i & 1 \end{pmatrix} \mathbf{T} \begin{pmatrix} x_i \\ 1 \end{pmatrix} + 2\mathbf{V}^t \begin{pmatrix} x_i \\ 1 \end{pmatrix} + C_{Ah} \quad (230)$$

where

$$\mathbf{S} = \begin{pmatrix} P_{xx} + 2P_{xy}C + P_{yy}C^2 & P_{xy}D + P_{yy}CD \\ P_{xy}D + P_{yy}CD & P_{yy}D^2 \end{pmatrix} \quad (231)$$

$$\mathbf{T} = \begin{pmatrix} Q_{xx}\lambda^2 + 2Q_{xy}\lambda E + Q_{yy}E^2 & Q_{xx}\lambda H + Q_{xy}[\lambda F + EH] + Q_{yy}EF \\ Q_{xx}\lambda H + Q_{xy}[\lambda F + EH] + Q_{yy}EF & Q_{xx}H^2 + 2Q_{xy}FH + Q_{yy}F^2 \end{pmatrix} \quad (232)$$

$$\mathbf{U}^t = \begin{pmatrix} K_x + K_y C & K_y D \end{pmatrix} \quad (233)$$

$$\mathbf{V}^t = \begin{pmatrix} L_x \lambda + L_y E & L_x H + L_y F \end{pmatrix} \quad (234)$$

Then we can write

$$\frac{A_{Mh}^2 + A_{Ah}^2}{\overline{A}x_i^2 + \overline{B}x_i + \overline{C}} = (S_{xx} + T_{xx})x_i^2 + 2(S_{xy} + U_x + T_{xy} + V_x)x_i + (S_{yy} + 2U_y + C_{Mh} + T_{yy} + 2V_y + C_{Ah}) = \quad (235)$$

The integration over this remaining degree of freedom is standard, so that

$$\int dx_i \exp\left(-\frac{1}{2}\{A_{Mh}^2 + A_{Ah}^2\}\right) = \sqrt{\frac{2\pi}{\overline{A}}} \exp\left(-\frac{1}{2}\left\{\overline{C} - \frac{\overline{B}^2}{4\overline{A}}\right\}\right) \quad (236)$$

It is the dependence on the deviation variables $\Delta\mathbf{Q}$ and ΔE through D , F and H , that is important, so we write $\overline{A}$, $\overline{B}$ and $\overline{C}$ in terms of them,

$$\overline{A} = P_{xx} + 2P_{xy}C + P_{yy}C^2 + Q_{xx}\lambda^2 + 2Q_{xy}\lambda E + Q_{yy}E^2 \quad (237)$$

$$\begin{aligned} \frac{1}{2}\overline{B} &= P_{xy}D + P_{yy}CD + Q_{xx}\lambda H + Q_{xy}[\lambda F + EH] + Q_{yy}EF + K_x + K_yC + L_x\lambda + L_yE \\ &= (K_x + K_yC + L_x\lambda + L_yE) + \begin{pmatrix} D & F & H \end{pmatrix} \begin{pmatrix} P_{xy} + P_{yy}C \\ Q_{xy}\lambda + Q_{yy}E \\ Q_{xx}\lambda + Q_{xy}E \end{pmatrix} = \overline{B}_0 + \mathbf{DFH}^t \bullet \overline{\mathbf{B}} \end{aligned} \quad (238)$$

$$\begin{aligned} \overline{C} &= P_{yy}D^2 + 2K_yD + C_{Mh} + Q_{xx}H^2 + 2Q_{xy}FH + Q_{yy}F^2 + 2(L_xH + L_yF) + C_{Ah} \\ &= (C_{Mh} + C_{Ah}) + \begin{pmatrix} D & F & H \end{pmatrix} \begin{pmatrix} P_{yy} & 0 & 0 \\ 0 & Q_{yy} & Q_{xy} \\ 0 & Q_{xy} & Q_{xx} \end{pmatrix} \begin{pmatrix} D \\ F \\ H \end{pmatrix} + 2 \begin{pmatrix} D & F & H \end{pmatrix} \begin{pmatrix} K_y \\ L_y \\ L_x \end{pmatrix} \\ &= \overline{C}_0 + 2\mathbf{DFH}^t \bullet \overline{\mathbf{C}} + \mathbf{DFH}^t \bullet \overline{\mathbf{C}}_2 \bullet \mathbf{DFH} \end{aligned} \quad (239)$$

Now

$$\frac{1}{4}\overline{B}^2 = \overline{B}_0^2 + 2\overline{B}_0\mathbf{DFH}^t \bullet \overline{\mathbf{B}} + \mathbf{DFH}^t \bullet (\overline{\mathbf{B}} \bullet \overline{\mathbf{B}}^t) \bullet \mathbf{DFH} \quad (240)$$

so that

$$\overline{C} - \frac{\overline{B}^2}{4\overline{A}} = \left(\overline{C}_0 - \frac{\overline{B}_0^2}{\overline{A}} \right) + 2 \left(\overline{C}^t - \frac{\overline{B}_0}{\overline{A}} \overline{B}^t \right) \bullet \mathbf{DFH} + \mathbf{DFH}^t \bullet \left(\overline{C}_2 - \frac{\overline{B} \bullet \overline{B}^t}{\overline{A}} \right) \bullet \mathbf{DFH} \quad (241)$$

Using $\mathbf{DFH} = \overline{\alpha} \bullet \Delta$ this becomes,

$$\begin{aligned} \overline{C} - \frac{\overline{B}^2}{4\overline{A}} &= \left(\overline{C}_0 - \frac{\overline{B}_0^2}{\overline{A}} \right) + 2 \left(\overline{C}^t - \frac{\overline{B}_0}{\overline{A}} \overline{B}^t \right) \bullet \overline{\alpha} \bullet \Delta + \Delta^t \bullet \overline{\alpha}^t \bullet \left(\overline{C}_2 - \frac{\overline{B} \bullet \overline{B}^t}{\overline{A}} \right) \bullet \overline{\alpha} \bullet \Delta \\ &= \overline{C}_{00} + \overline{C}_\alpha^t \bullet \Delta + \Delta^t \bullet \overline{C}_{2\alpha} \bullet \Delta \end{aligned} \quad (242)$$

The linear term produces an offset so we take $\Delta = \overline{\Delta} + \Delta_0$ and then choose Δ_0 to get rid of the linear terms,

$$\begin{aligned} &\overline{C}_{00} + \overline{C}_\alpha^t \bullet (\overline{\Delta} + \Delta_0) + (\overline{\Delta} + \Delta_0)^t \bullet \overline{C}_{2\alpha} \bullet (\overline{\Delta} + \Delta_0) = \\ &\overline{C}_{00} + \overline{C}_\alpha^t \bullet \Delta_0 + \Delta_0^t \bullet \overline{C}_{2\alpha} \bullet \Delta_0 + \left(\overline{C}_\alpha^t + 2\Delta_0^t \bullet \overline{C}_{2\alpha} \right) \bullet \overline{\Delta} + \overline{\Delta}^t \bullet \overline{C}_{2\alpha} \bullet \overline{\Delta} \end{aligned} \quad (243)$$

where we have used the fact that $\overline{C}_{2\alpha}$ is a symmetric matrix. Then identifying $\overline{C}_{2\alpha}$ as the resolution correlation matrix, $\mathbf{M}$, and taking the value of Δ_0 to be

$$\Delta_0 = -\frac{1}{2} \mathbf{M}^{-1} \bullet \overline{C}_\alpha \quad (244)$$

where

$$\overline{C}_\alpha^t = 2 \left(\overline{C}^t - \frac{\overline{B}_0}{\overline{A}} \overline{B}^t \right) \bullet \overline{\alpha} \quad (245)$$

$$\overline{C} = \begin{pmatrix} K_y \\ L_y \\ L_x \end{pmatrix} = \begin{pmatrix} a_2 \hat{\theta}_m + a_8 \hat{\theta}_0 - \tilde{a}_8 \hat{\gamma}_{00} - \tilde{a}_3 \hat{\gamma}_{10} \\ a_6 \hat{\theta}_a + a_{10} \hat{\theta}_3 - \tilde{a}_{10} \hat{\gamma}_{30} - \tilde{a}_4 \hat{\gamma}_{20} \\ a_5 \hat{\theta}_a + a_9 \hat{\theta}_3 - \tilde{a}_9 \hat{\gamma}_{30} \end{pmatrix} \quad (246)$$

$$\overline{B} = \begin{pmatrix} P_{xy} + P_{yy}C \\ Q_{xy}\lambda + Q_{yy}E \\ Q_{xx}\lambda + Q_{xy}E \end{pmatrix} = \begin{pmatrix} \tilde{b}_0 + \tilde{b}_1 C \\ \tilde{b}_4 \lambda + \tilde{b}_2 E \\ \tilde{b}_3 \lambda + \tilde{b}_4 E \end{pmatrix} \quad (247)$$

$$\begin{aligned}\overline{B}_0 &= K_x + K_y C + L_x \lambda + L_y E = \\ & a_1 \widehat{\theta}_m + a_7 \widehat{\theta}_0 - \widetilde{a}_7 \widehat{\gamma}_{00} + \left(a_2 \widehat{\theta}_m + a_8 \widehat{\theta}_0 - \widetilde{a}_8 \widehat{\gamma}_{00} - \widetilde{a}_3 \widehat{\gamma}_{10} \right) C + \\ & \left(a_5 \widehat{\theta}_a + a_9 \widehat{\theta}_3 - \widetilde{a}_9 \widehat{\gamma}_{30} \right) \lambda + \left(a_6 \widehat{\theta}_a + a_{10} \widehat{\theta}_3 - \widetilde{a}_{10} \widehat{\gamma}_{30} - \widetilde{a}_4 \widehat{\gamma}_{20} \right) E\end{aligned}\quad (248)$$

and

$$\overline{C} - \frac{\overline{B}^2}{4A} = \overline{C}_{00} - \frac{1}{4} \overline{C}_\alpha^t \bullet \mathbf{M}^{-1} \bullet \overline{C}_\alpha + (\Delta - \Delta_0)^t \bullet \mathbf{M} \bullet (\Delta - \Delta_0) \quad (249)$$

where we have used the fact that the inverse of a symmetric matrix is itself symmetric. That is

$$\mathbf{M}^{-1} = |\mathbf{M}|^{-1} \text{adj } \mathbf{M} \quad (250)$$

and

$$(\text{adj } \mathbf{M})_{ij} = (-1)^{i+j} |\mathbf{M}(j \mid i)| \quad (251)$$

where $|\mathbf{M}(j \mid i)|$ is the determinant of $\mathbf{M}$ after removing the j th row and i th column, so that clearly $(\text{adj } \mathbf{M})_{ij} = (\text{adj } \mathbf{M})_{ji}$ for symmetric $\mathbf{M}$. Specifically

$$\mathbf{M} = \overline{\alpha}^t \bullet \left(\overline{C}_2 - \frac{\overline{\mathbf{B}} \bullet \overline{\mathbf{B}}^t}{\overline{A}} \right) \bullet \overline{\alpha} = \overline{\alpha}^t \bullet M \bullet \overline{\alpha} \quad (252)$$

where

$$M = \begin{pmatrix} P_{yy} & 0 & 0 \\ 0 & Q_{yy} & Q_{xy} \\ 0 & Q_{xy} & Q_{xx} \end{pmatrix} - \frac{1}{\overline{A}} \begin{pmatrix} \overline{\mathbf{B}}_1^2 & \overline{\mathbf{B}}_1 \overline{\mathbf{B}}_2 & \overline{\mathbf{B}}_1 \overline{\mathbf{B}}_3 \\ \overline{\mathbf{B}}_2 \overline{\mathbf{B}}_1 & \overline{\mathbf{B}}_2^2 & \overline{\mathbf{B}}_2 \overline{\mathbf{B}}_3 \\ \overline{\mathbf{B}}_3 \overline{\mathbf{B}}_1 & \overline{\mathbf{B}}_3 \overline{\mathbf{B}}_2 & \overline{\mathbf{B}}_3^2 \end{pmatrix} \quad (253)$$

Recall that

$$\overline{\mathbf{B}} = \begin{pmatrix} P_{xy} + P_{yy} C \\ Q_{xy} \lambda + Q_{yy} E \\ Q_{xx} \lambda + Q_{xy} E \end{pmatrix} \quad (254)$$

$$\begin{pmatrix} P_{xx} & P_{xy} \\ P_{yx} & P_{yy} \end{pmatrix} = \begin{pmatrix} \tilde{b}_5 & \tilde{b}_0 \\ \tilde{b}_0 & \tilde{b}_1 \end{pmatrix} \quad (255)$$

$$\begin{pmatrix} Q_{xx} & Q_{xy} \\ Q_{yx} & Q_{yy} \end{pmatrix} = \begin{pmatrix} \tilde{b}_3 & \tilde{b}_4 \\ \tilde{b}_4 & \tilde{b}_2 \end{pmatrix} \quad (256)$$

$$\overline{A} = P_{xx} + 2P_{xy}C + P_{yy}C^2 + Q_{xx}\lambda^2 + 2Q_{xy}\lambda E + Q_{yy}E^2 = \tilde{b}_5 + 2\tilde{b}_0C + \tilde{b}_1C^2 + \tilde{b}_3\lambda^2 + 2\tilde{b}_4\lambda E + \tilde{b}_2E^2 \quad (257)$$

Substituting these back into equation 251,

$$\mathbf{M} = \frac{1}{\overline{A}} \overline{\alpha}^t \bullet \begin{pmatrix} \overline{A}\tilde{b}_1 - \tilde{b}_{01}^2 & -\tilde{b}_{01}\tilde{b}_{24} & -\tilde{b}_{01}\tilde{b}_{34} \\ -\tilde{b}_{01}\tilde{b}_{24} & \overline{A}\tilde{b}_2 - \tilde{b}_{24}^2 & \overline{A}\tilde{b}_4 - \tilde{b}_{24}\tilde{b}_{34} \\ -\tilde{b}_{01}\tilde{b}_{34} & \overline{A}\tilde{b}_4 - \tilde{b}_{24}\tilde{b}_{34} & \overline{A}\tilde{b}_3 - \tilde{b}_{34}^2 \end{pmatrix} \bullet \overline{\alpha} \quad (258)$$

where

$$\tilde{b}_{01} = \tilde{b}_0 + \tilde{b}_1C \quad (259)$$

$$\tilde{b}_{24} = \tilde{b}_2E + \tilde{b}_4\lambda \quad (260)$$

$$\tilde{b}_{34} = \tilde{b}_3\lambda + \tilde{b}_4E \quad (261)$$

We will need the determinant of $\mathbf{M}$ in order to compute the normalization of the TPF. To this end, note that

$$|\overline{\alpha}| = |\overline{\alpha}^t| = h_e (d_x f_y - d_y f_x) = (2D_n k_F \sin(2\theta_S))^{-1} \quad (262)$$

so that

$$|\mathbf{M}| = h_e^2 (d_x f_y - d_y f_x)^2 \left(\tilde{b}_2\tilde{b}_3 - \tilde{b}_4^2 \right) \left(\tilde{b}_1\tilde{b}_5 - \tilde{b}_0^2 \right) \frac{1}{\overline{A}} \quad (263)$$

Thus we have the result that the horizontal resolution function looks like

$$R_h(\Delta - \Delta_0) = R_{h0} \exp \left(-\frac{1}{2} (\Delta - \Delta_0)^t \bullet \mathbf{M} \bullet (\Delta - \Delta_0) \right) \quad (264)$$

Let us examine the offset, Δ_0 , to convince ourselves that it makes sense. Take the case of elastic scattering with relaxed collimations except for the geometrical collimation from monochromator to sample and from sample to analyzer. Then

$$\mathbf{P} = \begin{pmatrix} a_1^2 & a_1 a_2 \\ a_1 a_2 & a_2^2 + \tilde{a}_3^2 \end{pmatrix} = \frac{1}{(\eta_M k_I)^2} \begin{pmatrix} \kappa_m^2 \tan^2(\omega_{me}) & \kappa_m \tan(\omega_{me}) \\ \kappa_m \tan(\omega_{me}) & 1 + (\eta_M/\alpha_{MS})^2 \end{pmatrix} \quad (265)$$

$$\mathbf{Q} = \begin{pmatrix} a_5^2 & a_5 a_6 \\ a_5 a_6 & a_6^2 + \tilde{a}_4^2 \end{pmatrix} = \frac{1}{(\eta_A k_F)^2} \begin{pmatrix} \kappa_a^2 \tan^2(\omega_{ae}) & -\kappa_a \tan(\omega_{ae}) \\ -\kappa_a \tan(\omega_{ae}) & 1 + (\eta_A/\alpha_{SA})^2 \end{pmatrix} \quad (266)$$

Set the monochromator deviations to zero and $\Delta\omega_a = 0$ $\vartheta_A = -\gamma_{20}$, so that $K_x = K_y = 0$ and

$$L_x = \frac{\gamma_{20} \tan(\omega_{ae})}{k_F \eta_A^2} \quad (267)$$

$$L_y = -\frac{\gamma_{20}}{k_F \eta_A^2} \left(1 + \frac{\eta_A^2}{\alpha_{SA}^2} \right) = -\frac{\gamma_{20} f}{k_F \eta_A^2} \quad (268)$$

This choice of deviation angles should preserve the elastic condition. At the elastic condition, $\lambda = 1$ and $C = -E = \tan(\theta_S)$. Then taking $\eta_M = \eta_A = \eta$, $\omega_{me} = \omega_{ae} = \omega$, $k_I = k_F = k$, $\alpha_{MS} = \alpha_{SA} = \alpha_G$ and $f = 1 + (\eta/\alpha_G)^2 = 1 + \varepsilon$

$$\overline{A} = \frac{2}{(\eta k)^2} (\tan^2(\omega) + 2 \tan(\omega) \tan(\theta_S) + f \tan^2(\theta_S)) \quad (269)$$

$$\overline{B} = \frac{1}{(\eta k)^2} \begin{pmatrix} \tan(\omega) + f \tan(\theta_S) \\ -\tan(\omega) - f \tan(\theta_S) \\ \tan^2(\omega) + \tan(\omega) \tan(\theta_S) \end{pmatrix} \quad (270)$$

$$\overline{B}_o = \frac{\gamma_{20}}{k \eta^2} (\tan(\omega) + f \tan(\theta_S)) \quad (271)$$

Also write $\tan(\omega) + \tan(\theta_S) = w$, so that

$$\overline{A} = \frac{2}{(\eta k)^2} (w^2 + \varepsilon \tan^2(\theta_S)) \quad (272)$$

$$\overline{\mathbf{B}} = \frac{1}{(\eta k)^2} \begin{pmatrix} w + \varepsilon \tan(\theta_S) \\ -w - \varepsilon \tan(\theta_S) \\ w \tan(\omega) \end{pmatrix} \quad (273)$$

$$\overline{B}_o = \frac{\gamma_{20}}{k\eta^2} (w + \varepsilon \tan(\theta_S)) \quad (274)$$

Then, setting $w + \varepsilon \tan(\theta_S) = g$, $2(w^2 + \varepsilon \tan^2(\theta_S)) = h^2$, $g/h = \hat{g}$ and $w/h = \hat{w}$, the resolution matrix is

$$\mathbf{M} = \frac{1}{(\eta k)^2} \overline{\alpha}^t \bullet \begin{pmatrix} f - \hat{g}^2 & \hat{g}^2 & -\hat{g}\hat{w} \tan(\omega) \\ \hat{g}^2 & f - \hat{g}^2 & (\hat{g}\hat{w} - 1) \tan(\omega) \\ -\hat{g}\hat{w} \tan(\omega) & (\hat{g}\hat{w} - 1) \tan(\omega) & (1 - \hat{w}^2) \tan^2(\omega) \end{pmatrix} \bullet \overline{\alpha} \quad (275)$$

In order to compute a simple case, set $w = 0$, which is the exact W-configuration of the spectrometer, so that

$$\mathbf{M} = \frac{1}{(\eta k)^2} \overline{\alpha}^t \bullet \begin{pmatrix} 1 + \frac{\varepsilon}{2} & \frac{\varepsilon}{2} & 0 \\ \frac{\varepsilon}{2} & 1 + \frac{\varepsilon}{2} & -\tan(\omega) \\ 0 & -\tan(\omega) & \tan^2(\omega) \end{pmatrix} \bullet \overline{\alpha} \quad (276)$$

Note that if $f = 1$, $\varepsilon = 0$ which means $\alpha_G \rightarrow \infty$, the resolution matrix is of reduced rank and determinant zero as we discussed previously. The determinant of the matrix in brackets is $\frac{\varepsilon}{2} \tan^2(\omega)$.

Recalling $|\overline{\alpha}| = (2D_n k_F \sin(2\theta_S))^{-1} = (2D_n k_F \alpha)^{-1} = r/\alpha$, gives

$$|\mathbf{M}| = \frac{\varepsilon}{32D_n^2 \eta^6 k^8} \cos^{-4}(\theta_S) \quad (277)$$

Then,

$$\mathbf{M}^{-1} = (\eta k)^2 \overline{\alpha}^{-1} \bullet \begin{pmatrix} 1 & -1 & \cot(\theta_S) \\ -1 & 1 + \frac{2}{\varepsilon} & -\left(1 + \frac{2}{\varepsilon}\right) \cot(\theta_S) \\ \cot(\theta_S) & -\left(1 + \frac{2}{\varepsilon}\right) \cot(\theta_S) & 2\left(1 + \frac{1}{\varepsilon}\right) \cot^2(\theta_S) \end{pmatrix} \bullet (\overline{\alpha}^t)^{-1} \quad (278)$$

where

$$\bar{\alpha}^{-1} = \frac{1}{|\bar{\alpha}|} \begin{pmatrix} h_e f_y & -h_e d_y & d_e f_y - d_y f_e \\ -h_e f_x & h_e d_x & d_x f_e - d_e f_x \\ 0 & 0 & d_x f_y - d_y f_x \end{pmatrix} = \begin{pmatrix} a & -A & B \\ -b & B & A \\ 0 & 0 & -1/r \end{pmatrix} \quad (279)$$

and $(\bar{\alpha}^t)^{-1} = (\bar{\alpha}^{-1})^t$. Also for $w = 0$

$$\bar{A} = \frac{2\varepsilon}{(\eta k)^2} \tan^2(\theta_S) \quad (280)$$

$$\bar{\mathbf{B}} = \frac{\varepsilon \tan(\theta_S)}{(\eta k)^2} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \quad (281)$$

$$\bar{B}_o = \frac{\varepsilon \tan(\theta_S)}{k\eta^2} \gamma_{20} \quad (282)$$

$$\bar{\mathbf{C}} = \begin{pmatrix} K_y \\ L_y \\ L_x \end{pmatrix} = -\frac{\gamma_{20}}{k\eta^2} \begin{pmatrix} 0 \\ f \\ \tan(\theta_S) \end{pmatrix} \quad (283)$$

$$\bar{\mathbf{C}}_\alpha = 2\bar{\alpha}^t \bullet \left(\bar{\mathbf{C}} - \frac{\bar{B}_0}{\bar{A}} \bar{\mathbf{B}} \right) = -2\frac{\gamma_{20}}{k\eta^2} \bar{\alpha}^t \bullet \begin{pmatrix} \frac{\varepsilon}{2} \\ 1 + \frac{\varepsilon}{2} \\ \tan(\theta_S) \end{pmatrix} \quad (284)$$

The equation for Δ_o was

$$\Delta_o = -\mathbf{M}^{-1} \bullet \bar{\alpha}^t \bullet \left(\bar{\mathbf{C}} - \frac{\bar{B}_0}{\bar{A}} \bar{\mathbf{B}} \right) = -\bar{\alpha}^{-1} \bullet M^{-1} \bullet \left(\bar{\mathbf{C}} - \frac{\bar{B}_0}{\bar{A}} \bar{\mathbf{B}} \right) \quad (285)$$

This simplifies to

$$\Delta_o = \frac{\gamma_{20}}{k\eta^2} (\eta k)^2 \bar{\alpha}^{-1} \bullet \begin{pmatrix} 1 & -1 & \cot(\theta_S) \\ -1 & 1 + \frac{2}{\varepsilon} & -\left(1 + \frac{2}{\varepsilon}\right) \cot(\theta_S) \\ \cot(\theta_S) & -\left(1 + \frac{2}{\varepsilon}\right) \cot(\theta_S) & 2\left(1 + \frac{1}{\varepsilon}\right) \cot^2(\theta_S) \end{pmatrix} \begin{pmatrix} \frac{\varepsilon}{2} \\ 1 + \frac{\varepsilon}{2} \\ \tan(\theta_S) \end{pmatrix} = \gamma_{20} k \begin{pmatrix} a & -A & B \\ -b & B & A \\ 0 & 0 & -1/r \end{pmatrix} \bullet \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \gamma_{20} k \begin{pmatrix} -A \\ B \\ 0 \end{pmatrix}$$

It is easy to show that this is exactly the change in the spectrometer setting, $\mathbf{Q}_o$, in Cooper and Nathans coordinates, that is calculated if the analyzer is

offset by γ_{20} so that $2\theta_S$ is increased by γ_{20} for the elastic condition, and the analyzer is set again for k . Note that in this case where we are in the exact W-configuration for elastic scattering, $\omega = -\theta_s$, the spectrometer offset is independent of the crystal mosaics and the geometrical collimation.

vertical resolution function of a triple-axis spectrometer

Recall

$$A_{Mv}^2 = \left(\frac{\delta_0}{\beta_0}\right)^2 + \left(\frac{\delta_0 - \delta_1 + 2\tilde{\chi}_m}{2\tilde{\eta}_{Mv}}\right)^2 + \left(\frac{\delta_0 - \delta_{00}}{\beta_{RM}}\right)^2 + \left(\frac{\delta_1}{\beta_1}\right)^2 + \left(\frac{\delta_1 - \delta_{10}}{\beta_{MS}}\right)^2 \quad (286)$$

$$A_{Av}^2 = \left(\frac{\delta_3}{\beta_3}\right)^2 + \left(\frac{\delta_3 - \delta_2 - 2\tilde{\chi}_a}{2\tilde{\eta}_{Av}}\right)^2 + \left(\frac{\delta_3 - \delta_{30}}{\beta_{AD}}\right)^2 + \left(\frac{\delta_2}{\beta_2}\right)^2 + \left(\frac{\delta_2 - \delta_{20}}{\beta_{SA}}\right)^2 \quad (287)$$

where

$$\tilde{\chi}_m = \sin |\omega_{me}| \chi_m \quad (288)$$

$$\tilde{\eta}_{Mv} = \eta_{Mv} \sin (\omega_{me}) \quad (289)$$

$$\tilde{\chi}_a = \sin |\omega_{ae}| \chi_a \quad (290)$$

$$\tilde{\eta}_{Av} = \eta_{Av} \sin (\omega_{ae}) \quad (291)$$

$$\delta_{10} = \delta_{00} + 2 \sin |\omega_{me}| \chi_m = \delta_{00} + 2\tilde{\chi}_m \quad (292)$$

$$\delta_{30} = \delta_{20} + 2 \sin |\omega_{ae}| \chi_a = \delta_{20} + 2\tilde{\chi}_a \quad (293)$$

We must integrate over the vertical degrees of freedom before the monochromator, δ_0 , and after the analyzer, δ_3 , in order to obtain the resolution function. So we write

$$A_{Mv}^2 = \delta_0^2 \left(\frac{1}{\beta_0^2} + \frac{1}{\beta_{RM}^2} + \frac{1}{4\tilde{\eta}_{Mv}^2} \right) - 2\delta_0 \left(\frac{\delta_1 - 2\tilde{\chi}_m}{4\tilde{\eta}_{Mv}^2} + \frac{\delta_{00}}{\beta_{RM}^2} \right) + \left(\frac{\delta_1 - 2\tilde{\chi}_m}{2\tilde{\eta}_{Mv}} \right)^2 + \left(\frac{\delta_{00}}{\beta_{RM}} \right)^2 + \left(\frac{\delta_1}{\beta_1} \right)^2 + \left(\frac{\delta_1 - \delta_{10}}{\beta_{MS}} \right)^2 \quad (294)$$

and define

$$\frac{1}{\hat{\beta}_M^2} = \frac{1}{\beta_0^2} + \frac{1}{\beta_{RM}^2} + \frac{1}{4\tilde{\eta}_{Mv}^2} \quad (295)$$

Then

$$\int d\delta_0 \exp \left(-\frac{1}{2} \left\{ \frac{1}{\hat{\beta}_M^2} \delta_0^2 - 2 \left(\frac{\delta_1 - 2\tilde{\chi}_m}{4\tilde{\eta}_{Mv}^2} + \frac{\delta_{00}}{\beta_{RM}^2} \right) \delta_0 \right\} \right) = \sqrt{2\pi} \hat{\beta}_M \exp \left(\frac{1}{2} \hat{\beta}_M^2 \left[\frac{\delta_1 - 2\tilde{\chi}_m}{4\tilde{\eta}_{Mv}^2} + \frac{\delta_{00}}{\beta_{RM}^2} \right]^2 \right) \quad (296)$$

so that the remaining δ_1 dependence for the monochromator vertical TPF becomes

$$P_{Mv} = N_M \sqrt{2\pi} \hat{\beta}_M \exp \left(-\frac{1}{2} C_{M\delta} \right) \exp \left(-\frac{1}{2} \{ A_{M\delta} \delta_1^2 - 2B_{M\delta} \delta_1 \} \right) \quad (297)$$

where we use equation 290 to write

$$C_{M\delta} = \frac{\delta_{00}^2}{\beta_{RM}^2} + \frac{\delta_{10}^2}{\beta_{MS}^2} + \frac{\tilde{\chi}_m^2}{\tilde{\eta}_{Mv}^2} - \hat{\beta}_M^2 \left[\frac{\delta_{00}}{\beta_{RM}^2} - \frac{\tilde{\chi}_m}{2\tilde{\eta}_{Mv}^2} \right]^2 = \frac{\delta_{00}^2}{\beta_{CM\delta}^2} + 4 \frac{\tilde{\chi}_m^2}{\beta_{CM\chi}^2} + 4 \frac{\delta_{00}\tilde{\chi}_m}{\beta_{CM\delta\chi}^2} \quad (298)$$

with

$$\frac{1}{\beta_{CM\delta}^2} = \frac{1}{\beta_{MS}^2} + \frac{1}{\beta_{RM}^2 + \beta_{0\eta}^2} = \frac{1}{\beta_{MS}^2} + \frac{\beta_0^2 + 4\tilde{\eta}_{Mv}^2}{\beta_{0\eta M}^4} \quad (299)$$

$$\frac{1}{\beta_{0\eta}^2} = \frac{1}{\beta_0^2} + \frac{1}{4\tilde{\eta}_{Mv}^2} \quad (300)$$

$$\beta_{0\eta M}^4 = \beta_0^2 \beta_{RM}^2 + (\beta_0^2 + \beta_{RM}^2) 4\tilde{\eta}_{Mv}^2 \quad (301)$$

$$\frac{1}{\beta_{CM\chi}^2} = \frac{1}{\beta_{MS}^2} + \frac{1}{4\tilde{\eta}_{Mv}^2 + \hat{\beta}_0^2} = \frac{1}{\beta_{MS}^2} + \frac{\beta_0^2 + \beta_{RM}^2}{\beta_{0\eta M}^4} \quad (302)$$

$$\frac{1}{\beta_{CM\delta\chi}^2} = \frac{1}{\beta_{MS}^2} + \frac{1}{\beta_{RM}^2 + 4\tilde{\eta}_{Mv}^2 r_0^2} = \frac{1}{\beta_{MS}^2} + \frac{\beta_0^2}{\beta_{0\eta M}^4} \quad (303)$$

$$r_0^2 = 1 + \frac{\beta_{RM}^2}{\beta_0^2} = \frac{\beta_{RM}^2}{\hat{\beta}_0^2} \quad (304)$$

$$\hat{\beta}_0^2 = \frac{\beta_0^2 \beta_{RM}^2}{\beta_0^2 + \beta_{RM}^2} \quad (305)$$

$$A_{M\delta} = \frac{1}{4\tilde{\eta}_{Mv}^2} + \frac{1}{\beta_1^2} + \frac{1}{\beta_{MS}^2} - \frac{\hat{\beta}_M^2}{16\tilde{\eta}_{Mv}^4} = \frac{1}{\beta_1^2} + \frac{1}{\beta_{MS}^2} + \frac{1}{4\tilde{\eta}_{Mv}^2 + \hat{\beta}_0^2} = \frac{1}{\beta_1^2} + \frac{1}{4\tilde{\eta}_{Mv}^2 + \hat{\beta}_0^2} = \frac{\hat{\beta}_0^2 + \hat{\beta}_1^2 + 4\tilde{\eta}_{Mv}^2}{\hat{\beta}_1^2 (4\tilde{\eta}_{Mv}^2 + \hat{\beta}_0^2)} \quad (306)$$

$$B_{M\delta} = \frac{\tilde{\chi}_m}{2\tilde{\eta}_{Mv}^2} + \frac{\delta_{10}}{\beta_{MS}^2} + \frac{\hat{\beta}_M^2}{4\tilde{\eta}_{Mv}^2} \left[\frac{\delta_{00}}{\beta_{RM}^2} - \frac{\tilde{\chi}_m}{2\tilde{\eta}_{Mv}^2} \right] = \frac{\tilde{\chi}_m}{2\tilde{\eta}_{Mv}^2} + \frac{\delta_{10}}{\beta_{MS}^2} + \frac{\hat{\beta}_0^2}{4\tilde{\eta}_{Mv}^2 + \hat{\beta}_0^2} \left[\frac{\delta_{00}}{\beta_{RM}^2} - \frac{\tilde{\chi}_m}{2\tilde{\eta}_{Mv}^2} \right] \quad (307)$$

We can use equation 290 once more to rewrite $B_{M\delta}$ as

$$B_{M\delta} = \frac{\delta_{00}}{\beta_{BM\delta}^2} + \frac{2\tilde{\chi}_m}{\beta_{BM\chi}^2} = \frac{\delta_{10}}{\beta_{MS}^2} + \frac{\delta_{00}/r_0^2 + 2\tilde{\chi}_m}{4\tilde{\eta}_{Mv}^2 + \hat{\beta}_0^2} \quad (308)$$

where

$$\frac{1}{\beta_{BM\delta}^2} = \frac{1}{\beta_{MS}^2} + \frac{\hat{\beta}_0^2/\beta_{RM}^2}{\hat{\beta}_0^2 + 4\tilde{\eta}_{Mv}^2} = \frac{1}{\beta_{CM\delta\chi}^2} \quad (309)$$

$$\frac{1}{\beta_{BM\chi}^2} = \frac{1}{\beta_{MS}^2} + \frac{1}{\hat{\beta}_0^2 + 4\tilde{\eta}_{Mv}^2} = \frac{1}{\beta_{CM\chi}^2} \quad (310)$$

The same procedure for the analyzer system yields,

$$A_{Av}^2 = \delta_3^2 \left(\frac{1}{\beta_3^2} + \frac{1}{\beta_{AD}^2} + \frac{1}{4\tilde{\eta}_{Av}^2} \right) - 2\delta_3 \left(\frac{\delta_2 + 2\tilde{\chi}_a}{4\tilde{\eta}_{Av}^2} + \frac{\delta_{30}}{\beta_{AD}^2} \right) + \left(\frac{\delta_2 + 2\tilde{\chi}_a}{2\tilde{\eta}_{Av}} \right)^2 + \left(\frac{\delta_{30}}{\beta_{AD}} \right)^2 + \left(\frac{\delta_2}{\beta_2} \right)^2 + \left(\frac{\delta_2 - \delta_{20}}{\beta_{SA}} \right)^2 \quad (311)$$

$$\frac{1}{\hat{\beta}_A^2} = \frac{1}{\beta_3^2} + \frac{1}{\beta_{AD}^2} + \frac{1}{4\tilde{\eta}_{Av}^2} \quad (312)$$

$$\int d\delta_3 \exp\left(-\frac{1}{2}\left\{\frac{1}{\beta_A^2}\delta_3^2 - 2\left(\frac{\delta_2+2\tilde{\chi}_a}{4\tilde{\eta}_{Av}^2} + \frac{\delta_{30}}{\beta_{AD}^2}\right)\delta_3\right\}\right) = \sqrt{2\pi}\hat{\beta}_A \exp\left(\frac{1}{2}\hat{\beta}_A^2\left[\frac{\delta_2+2\tilde{\chi}_a}{4\tilde{\eta}_{Av}^2} + \frac{\delta_{30}}{\beta_{AD}^2}\right]^2\right) \quad (313)$$

$$P_{Av} = N_A \sqrt{2\pi}\hat{\beta}_A \exp\left(-\frac{1}{2}C_{A\delta}\right) \exp\left(-\frac{1}{2}\{A_{A\delta}\delta_2^2 - 2B_{A\delta}\delta_2\}\right) \quad (314)$$

We can use equation 291 to write $C_{A\delta}$ as

$$C_{A\delta} = \frac{\delta_{30}^2}{\beta_{AD}^2} + \frac{\delta_{20}^2}{\beta_{SA}^2} + \frac{\tilde{\chi}_a^2}{\tilde{\eta}_{Av}^2} - \hat{\beta}_A^2 \left[\frac{\delta_{30}}{\beta_{AD}^2} + \frac{\tilde{\chi}_a}{2\tilde{\eta}_{Av}^2} \right]^2 = \frac{\delta_{30}^2}{\beta_{CA\delta}^2} + 4\frac{\tilde{\chi}_a^2}{\beta_{CA\chi}^2} - 4\frac{\delta_{30}\tilde{\chi}_a}{\beta_{CA\delta\chi}^2} \quad (315)$$

with

$$\frac{1}{\beta_{CA\delta}^2} = \frac{1}{\beta_{SA}^2} + \frac{1}{\beta_{AD}^2 + \beta_{3\eta}^2} = \frac{1}{\beta_{SA}^2} + \frac{\beta_3^2 + 4\tilde{\eta}_{Av}^2}{\beta_{0\eta A}^4} \quad (316)$$

$$\frac{1}{\beta_{3\eta}^2} = \frac{1}{\beta_3^2} + \frac{1}{4\tilde{\eta}_{Av}^2} \quad (317)$$

$$\beta_{0\eta A}^4 = \beta_3^2 \beta_{AD}^2 + (\beta_3^2 + \beta_{AD}^2) 4\tilde{\eta}_{Av}^2 \quad (318)$$

$$\frac{1}{\beta_{CA\chi}^2} = \frac{1}{\beta_{SA}^2} + \frac{1}{4\tilde{\eta}_{Av}^2 + \hat{\beta}_3^2} = \frac{1}{\beta_{SA}^2} + \frac{\beta_3^2 + \beta_{AD}^2}{\beta_{0\eta A}^4} \quad (319)$$

$$\frac{1}{\beta_{CA\delta\chi}^2} = \frac{1}{\beta_{SA}^2} + \frac{1}{\beta_{AD}^2 + 4\tilde{\eta}_{Av}^2 r_3^2} = \frac{1}{\beta_{SA}^2} + \frac{\beta_3^2}{\beta_{0\eta A}^4} \quad (320)$$

$$r_3^2 = 1 + \frac{\beta_{AD}^2}{\beta_3^2} \quad (321)$$

$$\hat{\beta}_3^2 = \frac{\beta_3^2 \beta_{AD}^2}{\beta_3^2 + \beta_{AD}^2} = \frac{\beta_{AD}^2}{r_3^2} \quad (322)$$

$$A_{A\delta} = \frac{1}{4\tilde{\eta}_{Av}^2} + \frac{1}{\beta_2^2} + \frac{1}{\beta_{SA}^2} - \frac{\hat{\beta}_A^2}{16\tilde{\eta}_{Av}^4} = \frac{1}{\beta_2^2} + \frac{1}{\beta_{SA}^2} + \frac{1}{4\tilde{\eta}_{Av}^2 + \hat{\beta}_3^2} = \frac{1}{\beta_2^2} + \frac{1}{4\tilde{\eta}_{Av}^2 + \hat{\beta}_3^2} = \frac{\hat{\beta}_2^2 + \hat{\beta}_3^2 + 4\tilde{\eta}_{Av}^2}{\hat{\beta}_2^2(\hat{\beta}_2^2 + \hat{\beta}_3^2)} \quad (323)$$

and where

$$B_{A\delta} = -\frac{\tilde{\chi}_a}{2\tilde{\eta}_{Av}^2} + \frac{\delta_{20}}{\beta_{SA}^2} + \frac{\hat{\beta}_A^2}{4\tilde{\eta}_{Av}^2} \left[\frac{\delta_{30}}{\beta_{AD}^2} + \frac{\tilde{\chi}_a}{2\tilde{\eta}_{Av}^2} \right] = -\frac{\tilde{\chi}_a}{2\tilde{\eta}_{Av}^2} + \frac{\delta_{20}}{\beta_{SA}^2} + \frac{\hat{\beta}_3^2}{4\tilde{\eta}_{Av}^2 + \hat{\beta}_3^2} \left[\frac{\delta_{30}}{\beta_{AD}^2} + \frac{\tilde{\chi}_a}{2\tilde{\eta}_{Av}^2} \right] \quad (324)$$

We can use equation 291 again to write this as

$$B_{A\delta} = \frac{\delta_{30}}{\beta_{A\delta}^2} - \frac{2\tilde{\chi}_a}{\beta_{A\chi}^2} = \frac{\delta_{20}}{\beta_{SA}^2} + \frac{\delta_{30}/r_3^2 - 2\tilde{\chi}_a}{4\tilde{\eta}_{Av}^2 + \hat{\beta}_3^2} \quad (325)$$

where

$$\frac{1}{\beta_{A\delta}^2} = \frac{1}{\beta_{SA}^2} + \frac{\hat{\beta}_3^2/\beta_{AD}^2}{\hat{\beta}_3^2 + 4\tilde{\eta}_{Av}^2} = \frac{1}{\beta_{SA}^2} + \frac{1}{\beta_{AD}^2 + 4\tilde{\eta}_{Av}^2 r_3^2} = \frac{1}{\beta_{CA\delta\chi}^2} \quad (326)$$

$$\frac{1}{\beta_{A\chi}^2} = \frac{1}{\beta_{SA}^2} + \frac{1}{\hat{\beta}_3^2 + 4\tilde{\eta}_{Av}^2} = \frac{1}{\beta_{CA\chi}^2} \quad (327)$$

Then to obtain the vertical resolution function, substitute using $\Delta Q_z = z_f - z_i = \delta_2 k_{Fe} - \delta_1 k_{Ie}$ so that $\delta_2 = \frac{\kappa_a}{\kappa_m} \lambda \delta_1 + \Delta Q_z / k_{Fe}$. Then the total vertical resolution function can be obtained from

$$P_{Mv} P_{Av} = 2\pi N_M N_A \hat{\beta}_M \hat{\beta}_A \exp\left(-\frac{1}{2} C_{M\delta}\right) \exp\left(-\frac{1}{2} \{A_{M\delta} \delta_1^2 - 2B_{M\delta} \delta_1\}\right) \exp\left(-\frac{1}{2} C_{A\delta}\right) \exp\left(-\frac{1}{2} \left\{A_{A\delta} \left[\frac{\kappa_a}{\kappa_m} \lambda \delta_1 + \Delta Q_z / k_{Fe}\right]^2 - 2B_{A\delta} \left[\frac{\kappa_a}{\kappa_m} \lambda \delta_1 + \Delta Q_z / k_{Fe}\right]\right\}\right) \quad (328)$$

Taking $\frac{\kappa_a}{\kappa_m} \lambda = \bar{\lambda} = \frac{k_{Ie}}{k_{Fe}}$, we write this as a quadratic in δ_1 since we must integrate away this last degree of freedom,

$$P_{Mv} P_{Av} = C_\delta \exp\left(-\frac{1}{2} \left\{ \left[A_{M\delta} + A_{A\delta} \bar{\lambda}^2 \right] \delta_1^2 - 2 \left[B_{M\delta} - A_{A\delta} \bar{\lambda} \Delta Q_z / k_{Fe} + B_{A\delta} \bar{\lambda} \right] \delta_1 \right\}\right) \exp\left(-\frac{1}{2} \left\{ A_{A\delta} [\Delta Q_z / k_{Fe}]^2 - 2B_{A\delta} [\Delta Q_z / k_{Fe}] \right\}\right) \quad (329)$$

where

$$C_\delta = 2\pi N_M N_A \hat{\beta}_M \hat{\beta}_A \exp\left(-\frac{1}{2} C_{M\delta}\right) \exp\left(-\frac{1}{2} C_{A\delta}\right) \quad (330)$$

Then integrating over δ_1

$$R_v = R_{0v} \int d\delta_1 P_{Mv} P_{Av} = R_{0v} C_\delta \exp \left(-\frac{1}{2} \left\{ A_{A\delta} [\Delta Q_z / k_{Fe}]^2 - 2B_{A\delta} [\Delta Q_z / k_{Fe}] \right\} \right) \sqrt{2\pi} \left[A_{M\delta} + A_{A\delta} \bar{\lambda}^2 \right]^{-1/2} \exp \left(\frac{1}{2} \frac{[B_{M\delta} - A_{A\delta} \bar{\lambda} \Delta Q_z / k_{Fe} + B_{A\delta} \bar{\lambda}]^2}{[A_{M\delta} + A_{A\delta} \bar{\lambda}^2]} \right) \quad (331)$$

and collecting the ΔQ_z terms

$$R_v = R_{0v} C_\delta \sqrt{2\pi} \frac{1}{k_{Ie}} [\bar{A}_{M\delta} + \bar{A}_{A\delta}]^{-1/2} \exp \left(\frac{1}{2} \frac{[\bar{B}_{M\delta} + \bar{B}_{A\delta}]^2}{\bar{A}_{M\delta} + \bar{A}_{A\delta}} \right) \exp \left(-\frac{1}{2} \left\{ \frac{\bar{A}_{A\delta} \bar{A}_{M\delta}}{\bar{A}_{M\delta} + \bar{A}_{A\delta}} \Delta Q_z^2 - 2 \frac{\bar{A}_{M\delta} \bar{B}_{A\delta} - \bar{A}_{A\delta} \bar{B}_{M\delta}}{\bar{A}_{M\delta} + \bar{A}_{A\delta}} \Delta Q_z \right\} \right) \quad (332)$$

where

$$\bar{A}_{M\delta} = A_{M\delta} / k_{Ie}^2 \quad (333)$$

$$\bar{A}_{A\delta} = A_{A\delta} / k_{Fe}^2 \quad (334)$$

$$\bar{B}_{M\delta} = B_{M\delta} / k_{Ie} \quad (335)$$

$$\bar{B}_{A\delta} = B_{A\delta} / k_{Fe} \quad (336)$$

Complete the square to find

$$R_v = R_{0v} C_\delta \sqrt{2\pi} \frac{1}{k_{Ie}} [\bar{A}_{M\delta} + \bar{A}_{A\delta}]^{-1/2} \exp \left(\frac{1}{2} \frac{[\bar{A}_{M\delta} \bar{B}_{A\delta} - \bar{A}_{A\delta} \bar{B}_{M\delta}]^2 / [\bar{A}_{M\delta} \bar{A}_{A\delta}]}{\bar{A}_{M\delta} + \bar{A}_{A\delta}} \right) \exp \left(\frac{1}{2} \frac{[\bar{B}_{M\delta} + \bar{B}_{A\delta}]^2}{\bar{A}_{M\delta} + \bar{A}_{A\delta}} \right) \exp \left(-\frac{1}{2} \frac{\bar{A}_{A\delta} \bar{A}_{M\delta}}{\bar{A}_{M\delta} + \bar{A}_{A\delta}} \left[\Delta Q_z - \frac{\bar{A}_{M\delta} \bar{B}_{A\delta} - \bar{A}_{A\delta} \bar{B}_{M\delta}}{\bar{A}_{M\delta} \bar{A}_{A\delta}} \right]^2 \right) \quad (337)$$

Thus we finally have that the vertical part of the resolution function looks like

$$R_v (\Delta_z - \Delta_{0z}) = R_{v0} \exp \left(-\frac{1}{2} M_{zz} (\Delta_z - \Delta_{0z})^2 \right) \quad (338)$$

with

$$M_{zz} = \frac{\overline{A}_{A\delta}\overline{A}_{M\delta}}{\overline{A}_{M\delta} + \overline{A}_{A\delta}} \quad (339)$$

and

$$\Delta_{0z} = \frac{\overline{A}_{M\delta}\overline{B}_{A\delta} - \overline{A}_{A\delta}\overline{B}_{M\delta}}{\overline{A}_{M\delta}\overline{A}_{A\delta}} \quad (340)$$

elastic resolution function for a two-axis spectrometer

When the analyzer is removed, the horizontal resolution function can be calculated as before by taking $\eta_{Ah} = \infty$ and $\theta_A = 0$. This means that $a_5 = a_6 = a_9 = \tilde{a}_9 = 0$, and then $b_3 = \tilde{b}_4 = 0$. This leaves no terms in the horizontal TPF that depend on x_2 . The resulting resolution matrix elements that correlate ΔE will be very small and there is thus no energy resolution. We thus have a two dimensional resolution matrix, but the normalization will be different than in the three axis case. It only makes sense in this case to treat the elastic scattering case, since even the two dimensional resolution matrix will depend on k_F . We also must be careful with any geometrical collimation terms having to do with the analyzer. For the vertical resolution function we can again take $\eta_{Av} = \infty$. In addition, there is no vertical deviation angle after the analyzer to integrate over and there are no deviation offsets after the sample. Then

$$A_{Av2}^2 = \left(\frac{\delta_2}{\beta_2}\right)^2 + \left(\frac{\delta_2}{\beta_{SD}}\right)^2 = \left(\frac{\delta_2}{\hat{\beta}_2}\right)^2 \quad (341)$$

Here β_2 is the effective vertical collimation between the sample and detector, and β_{SD} is any geometrical size collimation from the sample to detector. This means that the full vertical TPF is

$$P_{Mv}P_{Av} = \sqrt{2\pi}N_M\hat{\beta}_M \exp\left(-\frac{1}{2}C_{M\delta}\right) \exp\left(-\frac{1}{2}\{A_{M\delta}\delta_1^2 - 2B_{M\delta}\delta_1\}\right) \exp\left(-\frac{1}{2}\{A_{A\delta 2}[\delta_1 + \Delta Q_z/k_I]^2\}\right) \quad (342)$$

so that

$$A_{A\delta 2} = \frac{1}{\widehat{\beta_2^2}} \quad (343)$$

and

$$B_{A\delta} = 0 \quad (344)$$

Then we can use the equations 337 and 338 to describe the two-axis vertical resolution function as well.

triple-axis normalization

The normalization of the resolution function is required in order to compute counting rates from a convolution with some cross-section for the scattering from the sample,

$$I_R (\text{neutrons/sec}) = R_0 \int \exp \left(-\frac{1}{2} (\Delta - \Delta_0)^t \bullet \mathbf{M} \bullet (\Delta - \Delta_0) \right) CS (\text{barns}/[meV]) d\Delta \quad (345)$$

where we are using our favorite choice of units for the cross-section, and the meV is in square brackets to indicate that it is absent in the two-axis case, where also $d\Delta$ would not include dE . We have included the vertical resolution function into this expression as well, so that for triple-axis, $d\Delta$ represents a 4-dimensional integral and for the two-axis case it represents a 3-dimensional integral. Note that in both cases, using $\text{\AA}^{-1}$ as units for ΔQ , R_{03} has dimensions $\text{neutrons/sec}/\text{\AA}^{-1}$ along with some conversion factor from barns to $\text{\AA}^2$.

We can also write an expression for the counting rate based directly on the TPF, where we can directly write down the normalization,

$$I_P (\text{neutrons/sec}) = \varepsilon_D f \varphi_k \left(\text{reactor flux}/\text{\AA}^{-1} \right) \int P(\Delta k_i, \gamma_i, \delta_i, [\Delta k_f], \gamma_f, \delta_f) CS \left(\text{barns}/\left[\text{\AA}^{-1}\right] \right) d\Delta k_i d\gamma_i d\delta_i [d\Delta k_f] d\gamma_f d\delta_f \quad (346)$$

Here ε_D is the detector efficiency, f represents the collimator intrinsic losses, and φ_k is the reactor source flux per unit wave-vector. The TPF is

$$P(\Delta k_i, \gamma_i, \delta_i, [\Delta k_f], \gamma_f, \delta_f) = P_{Mh} P_{Ah} P_{Mv} P_{Av} \quad (347)$$

where the factors are from equations 22, 29, 295 and 312, with the exponential arguments for the horizontal parts coming from equations 351 and 181. In order to compare 343 and 344, the cross-section should be set to unity. However, the units of the cross-section are different in the two expressions, so we convert $CS(\text{barns}/[\text{meV}])$ to $CS(\text{barns}/[\text{\AA}^{-1}])$ by

$$\begin{aligned} CS(\text{barns}/[\text{meV}]) &= CS(\text{barns}/[\text{\AA}^{-1}]) / \frac{dE}{dk_F} = \\ &CS(\text{barns}/[\text{\AA}^{-1}]) / (2D_n k_F) \end{aligned} \quad (348)$$

Then setting $CS(\text{barns}/[\text{\AA}^{-1}]) = 1\text{barn}/[\text{\AA}^{-1}]$, the convolution for I_R can be done as a standard integral,

$$I_P = I_R = R_0 (2\pi)^2 |\mathbf{M}|^{-1/2} / (2D_n k_F) \quad (349)$$

so that

$$R_0 = I_P (2\pi)^{-2} |\mathbf{M}|^{1/2} (2D_n k_F) \quad (350)$$

This is a somewhat different approach than used by [CandA], and allows the generalization to offset corrections. Then we need to calculate

$$\begin{aligned} I_P(\text{neutrons}/\text{sec}) &= \varepsilon_D f \varphi_k (\text{reactor flux}/\text{\AA}^{-1}) \\ &\int P_{Mh} P_{Ah} P_{Mv} P_{Av} d\Delta k_i d\gamma_i d\delta_i [d\Delta k_f] d\gamma_f d\delta_f \end{aligned} \quad (351)$$

Firstly,

$$\int P_{Mh} d\Delta k_i d\gamma_i = k_I \int \exp\left(-\frac{1}{2} A_{Mh}^2\right) d\overline{\Delta k_i} d\gamma_i \quad (352)$$

where

$$\begin{aligned} A_{Mh}^2 &= \left(\frac{\gamma_i + 2\overline{\Delta k_i} \tan(\omega_{me}) - 2\Delta\omega_m}{\alpha_0}\right)^2 + \left(\frac{\gamma_i + \overline{\Delta k_i} \tan(\omega_{me}) - \gamma_{10}}{\eta_M}\right)^2 + \left(\frac{\gamma_i}{\alpha_1}\right)^2 \\ &+ \left(\frac{\gamma_i + 2\overline{\Delta k_i} \tan(\omega_{me}) - \gamma_{10}}{\alpha_{RM}}\right)^2 + \left(\frac{\gamma_i - \gamma_{10}}{\alpha_{MS}}\right)^2 \end{aligned} \quad (353)$$

Rewriting this as a quadratic form in γ_i and $\overline{\Delta k_i}$

$$A_{Mh}^2 = A_{M\gamma\gamma}\gamma_i^2 + A_{Mkk}\overline{\Delta k_i}^2 + 2A_{M\gamma k}\gamma_i\overline{\Delta k_i} + B_{M\gamma}\gamma_i + B_{Mk}\overline{\Delta k_i} + C_M =$$

$$\left(\begin{array}{c} \gamma_i \\ \overline{\Delta k_i} \end{array} \right) \bullet \mathbf{A}_M \bullet \left(\begin{array}{c} \gamma_i \\ \overline{\Delta k_i} \end{array} \right) + \mathbf{B}_M \bullet \left(\begin{array}{c} \gamma_i \\ \overline{\Delta k_i} \end{array} \right) + C_M \quad (354)$$

where

$$A_{M\gamma\gamma} = \frac{1}{\alpha_0^2} + \frac{1}{\eta_M^2} + \frac{1}{\alpha_1^2} + \frac{1}{\alpha_{RM}^2} + \frac{1}{\alpha_{MS}^2} = \overline{A}_{M\gamma\gamma} \quad (355)$$

$$A_{Mkk} = \left(\frac{4}{\alpha_0^2} + \frac{1}{\eta_M^2} + \frac{4}{\alpha_{RM}^2} \right) \tan^2(\omega_{me}) = \overline{A}_{Mkk} \tan^2(\omega_{me}) \quad (356)$$

$$A_{M\gamma k} = \left(\frac{2}{\alpha_0^2} + \frac{1}{\eta_M^2} + \frac{2}{\alpha_{RM}^2} \right) \tan(\omega_{me}) = \overline{A}_{M\gamma k} \tan(\omega_{me}) \quad (357)$$

$$B_{M\gamma} = -2 \left(\frac{2\Delta\omega_m}{\alpha_0^2} + \frac{\gamma_{10}}{\eta_M^2} + \frac{\gamma_{10}}{\alpha_{RM}^2} + \frac{\gamma_{10}}{\alpha_{MS}^2} \right) = -2\overline{B}_{M\gamma} \quad (358)$$

$$B_{Mk} = -2 \left(\frac{4\Delta\omega_m}{\alpha_0^2} + \frac{\gamma_{10}}{\eta_M^2} + \frac{2\gamma_{10}}{\alpha_{RM}^2} \right) \tan(\omega_{me}) = -2\overline{B}_{Mk} \tan(\omega_{me}) \quad (359)$$

$$C_M = \frac{(\gamma_{00} + \Delta\omega_m)^2}{\eta_M^2} + \frac{\gamma_{00}^2}{\alpha_{RM}^2} + \frac{(\gamma_{00} + 2\Delta\omega_m)^2}{\alpha_{MS}^2} \quad (360)$$

We will use the integration formula

$$\int \exp \left(-\frac{1}{2} [\mathbf{X}^t \bullet \mathbf{A} \bullet \mathbf{X} + \mathbf{B}^t \bullet \mathbf{X}] \right) d\mathbf{X}^n = (2\pi)^{n/2} |\mathbf{A}|^{-1/2} \exp \left(\frac{1}{8} \mathbf{B}^t \bullet \mathbf{A}^{-1} \bullet \mathbf{B} \right) \quad (361)$$

with

$$|\mathbf{A}_M| = A_{M\gamma\gamma}A_{Mkk} - A_{M\gamma k}^2 =$$

$$\left[\frac{1}{\eta_M^2} \left(\frac{1}{\alpha_0^2} + \frac{1}{\alpha_1^2} \right) + 4 \frac{1}{\alpha_0^2} \frac{1}{\alpha_1^2} \right] \tan^2(\omega_{me}) = |\overline{\mathbf{A}}_M| \tan^2(\omega_{me}) \quad (362)$$

where

$$\frac{1}{\hat{\alpha}_0^2} = \frac{1}{\alpha_0^2} + \frac{1}{\alpha_{RM}^2} \quad (363)$$

$$\frac{1}{\hat{\alpha}_1^2} = \frac{1}{\alpha_1^2} + \frac{1}{\alpha_{MS}^2} \quad (364)$$

and

$$\mathbf{A}_M^{-1} = \begin{pmatrix} A_{Mkk} & -A_{M\gamma k} \\ -A_{M\gamma k} & A_{M\gamma\gamma} \end{pmatrix} / |\mathbf{A}_M| \quad (365)$$

to obtain

$$2\pi k_I |\mathbf{A}_M|^{-1/2} \exp \left(\frac{1}{2} \left[\tilde{A}_{M\gamma\gamma} \bar{B}_{Mk}^2 - 2\tilde{A}_{M\gamma k} \bar{B}_{M\gamma} \bar{B}_{Mk} + \tilde{A}_{Mkk} \bar{B}_{M\gamma}^2 \right] - \frac{1}{2} C_M \right) \quad (366)$$

with

$$\tilde{A}_{M\gamma\gamma} = \frac{\bar{A}_{M\gamma\gamma}}{|\bar{\mathbf{A}}_M|} = \frac{(\hat{\alpha}_0^2 + \hat{\alpha}_1^2) \eta_M^2 + \hat{\alpha}_0^2 \hat{\alpha}_1^2}{\hat{\alpha}_0^2 + \hat{\alpha}_1^2 + 4\eta_M^2} \quad (367)$$

$$\tilde{A}_{Mkk} = \frac{\bar{A}_{Mkk}}{|\bar{\mathbf{A}}_M|} = \frac{(\hat{\alpha}_0^2 + 4\eta_M^2) \hat{\alpha}_1^2}{\hat{\alpha}_0^2 + \hat{\alpha}_1^2 + 4\eta_M^2} \quad (368)$$

$$\tilde{A}_{M\gamma k} = \frac{\bar{A}_{M\gamma k}}{|\bar{\mathbf{A}}_M|} = \frac{(\hat{\alpha}_0^2 + 2\eta_M^2) \hat{\alpha}_1^2}{\hat{\alpha}_0^2 + \hat{\alpha}_1^2 + 4\eta_M^2} \quad (369)$$

Secondly,

$$\int P_{Ah} d\Delta k_f d\gamma_f = k_F \int \exp \left(-\frac{1}{2} A_{Ah}^2 \right) d\bar{\Delta k}_f d\gamma_f \quad (370)$$

where

$$A_{Ah}^2 = \left(\frac{\gamma_f - 2\bar{\Delta k}_f \tan(\omega_{ae})}{\alpha_3} \right)^2 + \left(\frac{\gamma_f - \bar{\Delta k}_f \tan(\omega_{ae}) - \gamma_{30} + \Delta\omega_a}{\eta_A} \right)^2 + \left(\frac{\gamma_f}{\alpha_2} \right)^2 \\ + \left(\frac{\gamma_f - 2\bar{\Delta k}_f \tan(\omega_{ae}) - \gamma_{30}}{\alpha_{AD}} \right)^2 + \left(\frac{\gamma_f - \gamma_{30} + 2\Delta\omega_a}{\alpha_{SA}} \right)^2 \quad (371)$$

Note that the $\tan(\omega_{ae})$ and $\Delta\omega$ terms have changed sign compared to the expression for A_{Mh}^2 in equation ??, so we can immediately write

$$A_{A\gamma\gamma} = \frac{1}{\alpha_3^2} + \frac{1}{\eta_A^2} + \frac{1}{\alpha_2^2} + \frac{1}{\alpha_{AD}^2} + \frac{1}{\alpha_{SA}^2} = \overline{A}_{A\gamma\gamma} \quad (372)$$

$$A_{Akk} = \left(\frac{4}{\alpha_3^2} + \frac{1}{\eta_A^2} + \frac{4}{\alpha_{AD}^2} \right) \tan^2(\omega_{ae}) = \overline{A}_{Akk} \tan^2(\omega_{ae}) \quad (373)$$

$$A_{A\gamma k} = - \left(\frac{2}{\alpha_3^2} + \frac{1}{\eta_A^2} + \frac{2}{\alpha_{AD}^2} \right) \tan(\omega_{ae}) = -\overline{A}_{A\gamma k} \tan(\omega_{ae}) \quad (374)$$

$$B_{A\gamma} = -2 \left(\frac{\gamma_{30} - \Delta\omega_a}{\eta_A^2} + \frac{\gamma_{30}}{\alpha_{AD}^2} + \frac{\gamma_{30} - 2\Delta\omega_a}{\alpha_{SA}^2} \right) = -2\overline{B}_{A\gamma} \quad (375)$$

$$B_{Ak} = 2 \left(\frac{\gamma_{30} - \Delta\omega_a}{\eta_A^2} + \frac{2\gamma_{30}}{\alpha_{AD}^2} \right) \tan(\omega_{ae}) = 2\overline{B}_{Ak} \tan(\omega_{ae}) \quad (376)$$

$$C_A = \frac{(\gamma_{30} - \Delta\omega_a)^2}{\eta_A^2} + \frac{\gamma_{30}^2}{\alpha_{AD}^2} + \frac{(\gamma_{30} - 2\Delta\omega_a)^2}{\alpha_{SA}^2} \quad (377)$$

$$|\mathbf{A}_A| = A_{A\gamma\gamma}A_{Akk} - A_{A\gamma k}^2 = \left[\frac{1}{\eta_A^2} \left(\frac{1}{\alpha_3^2} + \frac{1}{\alpha_2^2} \right) + 4\frac{1}{\alpha_3^2}\frac{1}{\alpha_2^2} \right] \tan^2(\omega_{ae}) = |\overline{\mathbf{A}}_A| \tan^2(\omega_{ae}) \quad (378)$$

where

$$\frac{1}{\widehat{\alpha}_3^2} = \frac{1}{\alpha_3^2} + \frac{1}{\alpha_{AD}^2} \quad (379)$$

$$\frac{1}{\widehat{\alpha}_2^2} = \frac{1}{\alpha_2^2} + \frac{1}{\alpha_{SA}^2} \quad (380)$$

and

$$\mathbf{A}_A^{-1} = \begin{pmatrix} A_{Akk} & -A_{A\gamma k} \\ -A_{A\gamma k} & A_{A\gamma\gamma} \end{pmatrix} / |\mathbf{A}_A| \quad (381)$$

with the result,

$$2\pi k_F |\mathbf{A}_A|^{-1/2} \exp \left(\frac{1}{2} \left[\tilde{A}_{A\gamma\gamma} \bar{B}_{Ak}^2 - 2\tilde{A}_{A\gamma k} \bar{B}_{A\gamma} \bar{B}_{Ak} + \tilde{A}_{Akk} \bar{B}_{A\gamma}^2 \right] - \frac{1}{2} C_A \right) \quad (382)$$

with

$$\tilde{A}_{A\gamma\gamma} = \frac{\bar{A}_{A\gamma\gamma}}{|\bar{\mathbf{A}}_A|} = \frac{(\hat{\alpha}_3^2 + \hat{\alpha}_2^2) \eta_A^2 + \hat{\alpha}_3^2 \hat{\alpha}_2^2}{\hat{\alpha}_3^2 + \hat{\alpha}_2^2 + 4\eta_A^2} \quad (383)$$

$$\tilde{A}_{Akk} = \frac{\bar{A}_{Akk}}{|\bar{\mathbf{A}}_A|} = \frac{(\hat{\alpha}_3^2 + 4\eta_A^2) \hat{\alpha}_2^2}{\hat{\alpha}_3^2 + \hat{\alpha}_2^2 + 4\eta_A^2} \quad (384)$$

$$\tilde{A}_{A\gamma k} = \frac{\bar{A}_{A\gamma k}}{|\bar{\mathbf{A}}_A|} = \frac{(\hat{\alpha}_3^2 + 2\eta_A^2) \hat{\alpha}_2^2}{\hat{\alpha}_3^2 + \hat{\alpha}_2^2 + 4\eta_A^2} \quad (385)$$

Thirdly, we can do the double integration of the monochromator vertical transmission probability, $N_M \exp(-\frac{1}{2} A_{Mv}^2)$, with A_{Mv}^2 given by 284,

$$\int d\delta_0 d\delta_1 \exp \left(-\frac{1}{2} A_{Mv}^2 \right) \quad (386)$$

$$A_{Mv}^2 = \left(\frac{\delta_0}{\beta_0} \right)^2 + \left(\frac{\delta_0 - \delta_1 + 2\tilde{\chi}_m}{2\tilde{\eta}_{Mv}} \right)^2 + \left(\frac{\delta_0 - \delta_{00}}{\beta_{RM}} \right)^2 + \left(\frac{\delta_1}{\beta_1} \right)^2 + \left(\frac{\delta_1 - \delta_{10}}{\beta_{MS}} \right)^2 \quad (387)$$

This is

$$\int d\delta_0 d\delta_1 \exp \left(-\frac{1}{2} \left\{ \frac{\delta_0^2}{\beta_{M0}^2} + \frac{\delta_1^2}{\beta_{M1}^2} - 2\delta_0 \delta_1 \left(\frac{1}{4\tilde{\eta}_{Mv}^2} \right) + \delta_0 B_{M\delta 0} + \delta_1 B_{M\delta 1} + C_{M\delta\chi} \right\} \right) \quad (388)$$

where

$$B_{M\delta 0} = \frac{\tilde{\chi}_m}{\tilde{\eta}_{Mv}^2} - 2 \frac{\delta_{00}}{\beta_{RM}^2} \quad (389)$$

$$B_{M\delta 1} = -\frac{\tilde{\chi}_m}{\tilde{\eta}_{Mv}^2} - 2 \frac{\delta_{10}}{\beta_{MS}^2} \quad (390)$$

$$\frac{1}{\beta_{M0}^2} = \frac{1}{\beta_0^2} + \frac{1}{\beta_{RM}^2} + \frac{1}{4\tilde{\eta}_{Mv}^2} = \frac{1}{\hat{\beta}_0^2} + \frac{1}{4\tilde{\eta}_{Mv}^2} \quad (391)$$

$$\frac{1}{\beta_{M1}^2} = \frac{1}{\beta_1^2} + \frac{1}{\beta_{MS}^2} + \frac{1}{4\tilde{\eta}_{Mv}^2} = \frac{1}{\widehat{\beta}_1^2} + \frac{1}{4\tilde{\eta}_{Mv}^2} \quad (392)$$

$$C_{M\delta\chi} = \frac{\tilde{\chi}_m^2}{\tilde{\eta}_{Mv}^2} + \frac{\delta_{00}^2}{\beta_{RM}^2} + \frac{\delta_{10}^2}{\beta_{MS}^2} = \frac{\tilde{\chi}_m^2}{\tilde{\eta}_{Mv}^2} + \frac{\delta_{00}^2}{\beta_{RM}^2} + \frac{(\delta_{00} + 2\tilde{\chi}_m)^2}{\beta_{MS}^2} \quad (393)$$

This can be rewritten

$$\int d\delta_M \exp \left(-\frac{1}{2} \{ \delta_M^t \bullet A_{M\delta} \bullet \delta_M + B_{M\delta} \bullet \delta_M + C_{M\delta\chi} \} \right) \quad (394)$$

Using again equation 359

$$\int d\delta_M \exp \left(-\frac{1}{2} \{ \delta_M^t \bullet A_{M\delta} \bullet \delta_M + B_{M\delta} \bullet \delta_M \} \right) = 2\pi |A_{M\delta}|^{-1/2} \exp \left(\frac{1}{8} B_{M\delta}^t \bullet A_{M\delta}^{-1} \bullet B_{M\delta} \right) \quad (395)$$

with

$$A_{M\delta} = \begin{pmatrix} \frac{1}{\beta_{M0}^2} & -\frac{1}{4\tilde{\eta}_{Mv}^2} \\ -\frac{1}{4\tilde{\eta}_{Mv}^2} & \frac{1}{\beta_{M1}^2} \end{pmatrix} \quad (396)$$

$$B_{M\delta}^t = \begin{pmatrix} \frac{\tilde{\chi}_m}{\tilde{\eta}_{Mv}^2} - 2\frac{\delta_{00}}{\beta_{RM}^2} & -\frac{\tilde{\chi}_m}{\tilde{\eta}_{Mv}^2} - 2\frac{\delta_{10}}{\beta_{MS}^2} \\ \frac{\tilde{\chi}_m}{\tilde{\eta}_{Mv}^2} - 2\frac{\delta_{00}}{\beta_{RM}^2} & \frac{\tilde{\chi}_m}{\tilde{\eta}_{Mv}^2} \left[-1 - \frac{4\tilde{\eta}_{Mv}^2}{\beta_{MS}^2} \right] - 2\frac{\delta_{00}}{\beta_{MS}^2} \end{pmatrix} = \begin{pmatrix} B_{M\delta 0} & B_{M\delta 1} \end{pmatrix} \quad (397)$$

$$|A_{M\delta}| = \frac{1}{\widehat{\beta}_0^2} \frac{1}{\widehat{\beta}_1^2} + \left(\frac{1}{\widehat{\beta}_0^2} + \frac{1}{\widehat{\beta}_1^2} \right) \frac{1}{4\tilde{\eta}_{Mv}^2} = \frac{1}{4\tilde{\eta}_{Mv}^2 \beta_M^2} \quad (398)$$

$$\beta_M = \frac{\widehat{\beta}_0 \widehat{\beta}_1}{\left(\widehat{\beta}_0^2 + \widehat{\beta}_1^2 + 4\tilde{\eta}_{Mv}^2 \right)^{1/2}} \quad (399)$$

$$A_{M\delta}^{-1} = \begin{pmatrix} \frac{1}{\beta_{M0}^2} & \frac{1}{4\tilde{\eta}_{Mv}^2} \\ \frac{1}{4\tilde{\eta}_{Mv}^2} & \frac{1}{\beta_{M1}^2} \end{pmatrix} / |A_{M\delta}| \quad (400)$$

Putting this all together, we get

$$\sqrt{2\pi}R_M\beta_M \exp\left(-\frac{1}{2}\left\{C_{M\delta\chi}-\beta_M^2\tilde{\eta}_{Mv}^2B_{M\delta 01}\right\}\right) \quad (401)$$

where

$$B_{M\delta 01} = \frac{B_{M\delta 0}^2}{\beta_{M0}^2} + 2\frac{B_{M\delta 0}B_{M\delta 1}}{4\tilde{\eta}_{Mv}^2} + \frac{B_{M\delta 1}^2}{\beta_{M1}^2} \quad (402)$$

The exponential factor represents losses due to the conflict between crystal vertical displacement, tilt setting, and vertical collimations. Note that the only correction for sample size is through the geometrical collimation angle, β_{MS} . The actual relative position of the sample and beam is not taken into account. This means that we assume the vertical displacement and tilt are set so that the beam is optimally directed to the sample.

Fourthly, we have the vertical transmission function for the analyzer system from $N_A \exp\left(-\frac{1}{2}A_{Av}^2\right)$, with A_{Av}^2 given by 285,

$$\int d\delta_3 d\delta_2 \exp\left(-\frac{1}{2}A_{Av}^2\right) \quad (403)$$

$$A_{Av}^2 = \left(\frac{\delta_3}{\beta_3}\right)^2 + \left(\frac{\delta_3 - \delta_2 - 2\tilde{\chi}_a}{2\tilde{\eta}_{Av}}\right)^2 + \left(\frac{\delta_3 - \delta_{30}}{\beta_{AD}}\right)^2 + \left(\frac{\delta_2}{\beta_2}\right)^2 + \left(\frac{\delta_2 - \delta_{20}}{\beta_{SA}}\right)^2 \quad (404)$$

This is the same as the monochromator system integral with $\delta_0 \rightarrow \delta_3$, $\delta_1 \rightarrow \delta_2$, $\delta_{00} \rightarrow \delta_{30}$, $\delta_{10} \rightarrow \delta_{20}$, $\beta_0 \rightarrow \beta_3$, $\beta_1 \rightarrow \beta_2$, $\beta_{RM} \rightarrow \beta_{AD}$, $\beta_{MS} \rightarrow \beta_{SA}$, $\tilde{\eta}_{Mv} \rightarrow \tilde{\eta}_{Av}$ and $\tilde{\chi}_m \rightarrow -\tilde{\chi}_a$ so that we get

$$\sqrt{2\pi}R_A\beta_A \exp\left(-\frac{1}{2}\left\{C_{A\delta\chi}-\beta_A^2\tilde{\eta}_{Av}^2B_{A\delta 32}\right\}\right) \quad (405)$$

with

$$B_{A\delta 32} = \frac{B_{A\delta 3}^2}{\beta_{A3}^2} + 2\frac{B_{A\delta 3}B_{A\delta 2}}{4\tilde{\eta}_{Av}^2} + \frac{B_{A\delta 2}^2}{\beta_{A2}^2} \quad (406)$$

$$\beta_A = \frac{\hat{\beta}_3\hat{\beta}_2}{\left(\hat{\beta}_3^2 + \hat{\beta}_2^2 + 4\tilde{\eta}_{Av}^2\right)^{1/2}} \quad (407)$$

$$C_{A\delta\chi} = \frac{\tilde{\chi}_a^2}{\tilde{\eta}_{Av}^2} + \frac{\delta_{30}^2}{\beta_{AD}^2} + \frac{\delta_{20}^2}{\beta_{SA}^2} = \frac{\tilde{\chi}_a^2}{\tilde{\eta}_{Av}^2} + \frac{\delta_{30}^2}{\beta_{AD}^2} + \frac{(\delta_{30} - 2\tilde{\chi}_a)^2}{\beta_{SA}^2} \quad (408)$$

$$\begin{pmatrix} B_{A\delta 3} & B_{A\delta 2} \end{pmatrix} = \begin{pmatrix} -\frac{\tilde{\chi}_a}{\tilde{\eta}_{Av}^2} - 2\frac{\delta_{30}}{\beta_{AD}^2} & \frac{\tilde{\chi}_a}{\tilde{\eta}_{Av}^2} - 2\frac{\delta_{20}}{\beta_{SA}^2} \end{pmatrix} \quad (409)$$

$$\frac{1}{\beta_{A3}^2} = \frac{1}{\beta_3^2} + \frac{1}{\beta_{AD}^2} + \frac{1}{4\tilde{\eta}_{Av}^2} = \frac{1}{\hat{\beta}_3^2} + \frac{1}{4\tilde{\eta}_{Av}^2} \quad (410)$$

$$\frac{1}{\beta_{A2}^2} = \frac{1}{\beta_2^2} + \frac{1}{\beta_{SA}^2} + \frac{1}{4\tilde{\eta}_{Av}^2} = \frac{1}{\hat{\beta}_2^2} + \frac{1}{4\tilde{\eta}_{Av}^2} \quad (411)$$

Combining the four results

$$\int P_{Mh} P_{Ah} P_{Mv} P_{Av} d\Delta k_i d\gamma_i d\delta_i [d\Delta k_f] d\gamma_f d\delta_f = (2\pi)^3 R_M R_A \beta_M^2 \beta_A^2 k_I k_F O_M O_A O_{Mv} O_{Av} |\mathbf{A}_M|^{-1/2} |\mathbf{A}_A|^{-1/2} \quad (412)$$

where

$$O_M = \exp \left(-\frac{1}{2} \left[C_M - \frac{1}{4} (A_{M\gamma\gamma} B_{Mk}^2 - 2A_{M\gamma k} B_{M\gamma} B_{Mk} + A_{Mkk} B_{M\gamma}^2) \right] \right) \quad (413)$$

$$O_A = \exp \left(-\frac{1}{2} \left[C_A - \frac{1}{4} (A_{A\gamma\gamma} B_{Ak}^2 - 2A_{A\gamma k} B_{A\gamma} B_{Ak} + A_{Akk} B_{A\gamma}^2) \right] \right) \quad (414)$$

$$O_{Mv} = \exp \left(-\frac{1}{2} [C_{M\delta\chi} - \beta_M^2 \tilde{\eta}_{Mv}^2 B_{M\delta 01}] \right) \quad (415)$$

$$O_{Av} = \exp \left(-\frac{1}{2} [C_{A\delta\chi} - \beta_A^2 \tilde{\eta}_{Av}^2 B_{A\delta 32}] \right) \quad (416)$$

Then the result for R_0 is

$$R_0 = 2\pi R_M R_A \varepsilon_D f \varphi_k \beta_M \beta_A k_I k_F O_M O_A O_{Mv} O_{Av} |\mathbf{A}_M|^{-1/2} |\mathbf{A}_A|^{-1/2} |\mathbf{M}|^{1/2} (2D_n k_F) \quad (417)$$

Recall

$$h_e (d_x f_y - d_y f_x) = (2D_n k_F \sin (2\theta_S))^{-1} \quad (418)$$

and also find that,

$$\tilde{b}_1 \tilde{b}_5 - \tilde{b}_0^2 = \frac{|\mathbf{A}_M|}{k_{Ie}^4} \quad (419)$$

$$\tilde{b}_2 \tilde{b}_3 - \tilde{b}_4^2 = \frac{|\mathbf{A}_A|}{k_{Fe}^4} \quad (420)$$

so that

$$|\mathbf{M}| = h_e^2 (d_x f_y - d_y f_x)^2 \left(\tilde{b}_2 \tilde{b}_3 - \tilde{b}_4^2 \right) \left(\tilde{b}_1 \tilde{b}_5 - \tilde{b}_0^2 \right) \frac{M_{zz}}{\bar{A}} = \quad (421)$$

$$(2D_n k_F \sin (2\theta_S))^{-2} \frac{|\mathbf{A}_M|}{k_{Ie}^4} \frac{|\mathbf{A}_A|}{k_{Fe}^4} \frac{M_{zz}}{\bar{A}}$$

Then

$$R_0 = \frac{1}{2\pi} \varepsilon_D f R_M R_A \frac{\varphi_k \lambda_{Ie} \lambda_{Fe}}{\sin (2\theta_S)} \beta_M \beta_A O_M O_A O_{Mv} O_{Av} \left(\frac{M_{zz}}{\bar{A}} \right)^{1/2} \quad (422)$$

where $\lambda_{Ie} = (2\pi) / k_{Ie}$ and $\lambda_{Fe} = (2\pi) / k_{Fe}$.

two-axis normalization

Following the triple-axis derivation

$$I_{P2} = I_{R2} = R_{02} (2\pi)^{3/2} |\mathbf{M}|^{-1/2} \quad (423)$$

The neutron path integrals for the analyzer are modified so that

$$\int P_{Ah} d\gamma_f = \sqrt{2\pi} \mathbf{A}_{A\gamma\gamma}^{-1/2} \quad (424)$$

$$\int P_{Av} d\delta_f = \sqrt{2\pi} \hat{\beta}_2 \quad (425)$$

$$\begin{aligned} & \int P_{Mh} P_{Ah} P_{Mv} P_{Av} d\Delta k_i d\gamma_i d\delta_i d\gamma_f d\delta_f = \\ & (2\pi)^3 N_M \widehat{\beta}_M A_{M\delta}^{-1/2} \widehat{\beta}_2 k_I O_M O_{Mv} |\mathbf{A}_M|^{-1/2} \mathbf{A}_{A\gamma\gamma}^{-1/2} = \\ & (2\pi)^{5/2} \beta_M \widehat{\beta}_2 k_I O_M O_{Mv} |\mathbf{A}_M|^{-1/2} \mathbf{A}_{A\gamma\gamma}^{-1/2} \end{aligned} \quad (426)$$

$$\begin{aligned} |\mathbf{M}| &= (d_x f_y - d_y f_x)^2 \widetilde{b}_2 \left(\widetilde{b}_1 \widetilde{b}_5 - \widetilde{b}_0^2 \right) \frac{M_{zz}}{A} = \\ & (\sin(2\theta_S))^{-2} \frac{|\mathbf{A}_M|}{k_{Ie}^4} \frac{\mathbf{A}_{A\gamma\gamma}}{k_{Fe}^2} \frac{M_{zz}}{A} \end{aligned} \quad (427)$$

Then

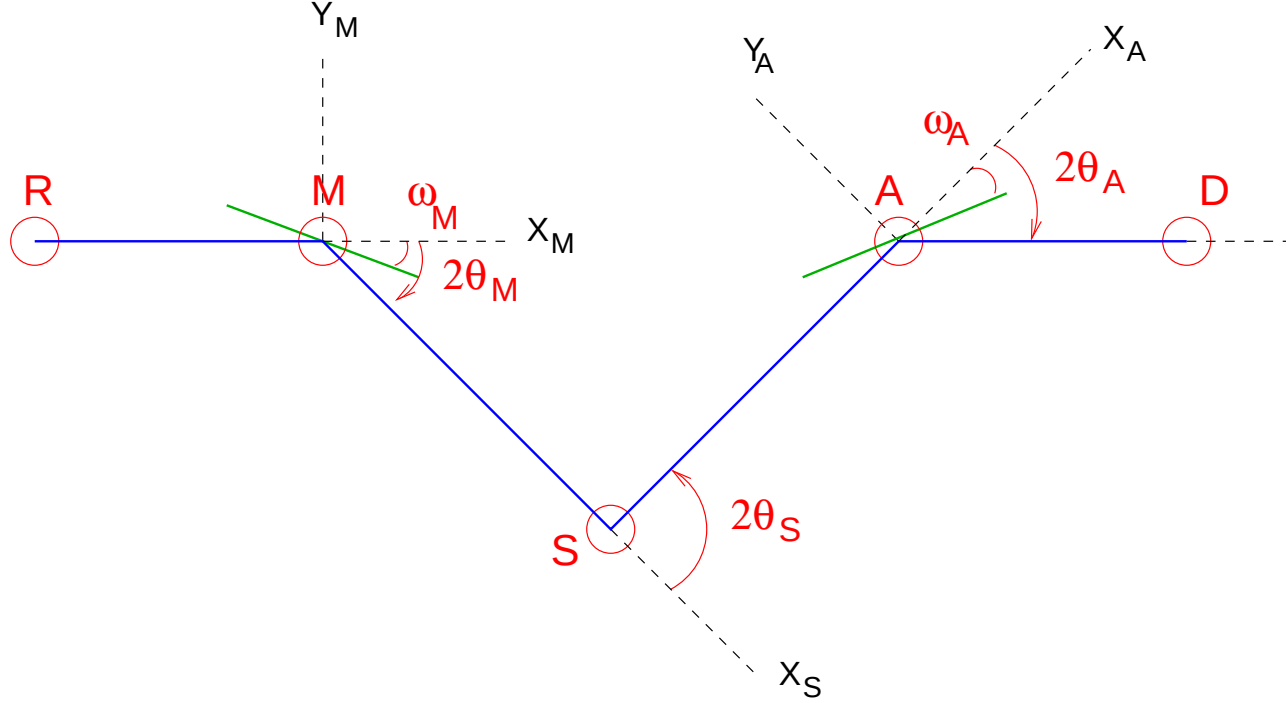
$$R_{02} = \frac{1}{2\pi} \varepsilon_D f R_M \frac{\varphi_k \lambda_{Ie} \lambda_{Fe}}{\sin(2\theta_S)} \beta_M \widehat{\beta}_2 O_M O_{Mv} \left(\frac{M_{zz}}{A} \right)^{1/2} \quad (428)$$

convolution procedure

A spectrometer must have nominal incoming and outgoing beam directions, which define a nominal scattering plane. Nominal incoming and outgoing wave-vectors must also be defined, and these are related to nominal reference rotation angles of the monochromator and analyzer crystals. There is also an initial crystal alignment which determines the reference reciprocal lattice vectors that lie in the scattering plane, which in turn defines the zero of the sample crystal rotation angle. Specifically, when the first reference reciprocal lattice vector is perpendicular to the nominal incident beam direction, then the sample crystal rotation angle is at zero. It is possible to rotate the goniometer arcs, so that some other reciprocal lattice vectors occupy the original alignment positions. The nominal wave-vector transfer, $\mathbf{Q}$, is defined in terms of the actual lattice vectors occupying the alignment positions, and the reciprocal lattice indices of $\mathbf{Q}$ should place it in the nominal scattering plane. While the resolution function of any monochromator and analyzer crystal pair can be specified with respect to the nominal $\mathbf{Q}$, the cross-section is specified in terms of reciprocal space. This is why it is important to know the alignment lattice vectors for any data point. The reciprocal lattice point corresponding to $\mathbf{Q}$, should lie in the nominal scattering plane, since the spectrometer setting is defined by the crystal rotation angle and the sample scattering angle.

crystal positions and angles

A discussion about the calculation of crystal displacement angles and geometrical size collimations follows.



The above diagram shows a triple-axis scattering plane with nominal positions for the monochromator, sample and analyzer crystals. When the monochromator or analyzer is a multocrystal, the individual components can be positioned away from the nominal position and the crystal setting angle can be different the nominal ω value also. Suppose for example that a monochromator element is positioned at $\mathbf{r}_M + \mathbf{d}_M$. This is at unit direction

$$\mathbf{u}_m = \sin(\delta_{00})\hat{\mathbf{z}} + \cos(\delta_{00}) [\sin(\gamma_{00})\hat{\mathbf{y}}_M + \cos(\gamma_{00})\hat{\mathbf{x}}_M] \quad (429)$$

where δ_{00} is the vertical deviation angle of the crystal element's position with positive above the scattering plane. and γ_{00} is the horizontal deviation angle of the crystal element's position with positive c.c.w. toward the y-axis (see the following figure). Note that the x and y-axes are completed as right handed coordinate systems with the z-axis up out of the scattering plane. The unit direction is also given by

$$\mathbf{u}_m = \frac{d_{Mz}}{D_M} \hat{\mathbf{z}} + \frac{d_{My}}{D_M} \hat{\mathbf{y}}_M + \frac{L_M + d_{Mx}}{D_M} \hat{\mathbf{x}}_M \quad (430)$$

where L_M is the separation distance of the nominal positions of reactor and monochromator, and D_M is the actual distance from the reactor to this crystal element as given by

$$D_M^2 = d_{Mz}^2 + d_{My}^2 + (L_M + d_{Mx})^2 \quad (431)$$

so we obtain the deviation position angles as

$$\sin(\delta_{00}) = \frac{d_{Mz}}{D_M} \quad (432)$$

$$\sin(\gamma_{00}) = \frac{d_{My}}{D_M \cos(\delta_{00})} \quad (433)$$

Now if the sample can be displaced as well to $\mathbf{r}_M + \mathbf{r}_S + \mathbf{d}_S$, then the unit direction from the monochromator element position to the sample is

$$\mathbf{u}_{ms} = (\mathbf{r}_S + \mathbf{d}_S - \mathbf{d}_M) / D_{MS} \quad (434)$$

where $D_{MS} = |\mathbf{r}_S + \mathbf{d}_S - \mathbf{d}_M|$. Now to put $\mathbf{d}_S$ and $\mathbf{d}_M$ in the same coordinate system

$$\hat{\mathbf{x}}_M = \cos(2\theta_M) \hat{\mathbf{x}}_S - \sin(2\theta_M) \hat{\mathbf{y}}_S \quad (435)$$

$$\hat{\mathbf{y}}_M = \sin(2\theta_M) \hat{\mathbf{x}}_S + \cos(2\theta_M) \hat{\mathbf{y}}_S \quad (436)$$

so that

$$\begin{aligned} \mathbf{u}_{ms} D_{MS} = & [L_S + d_{Sx} - d_{Mx} \cos(2\theta_M) - d_{My} \sin(2\theta_M)] \hat{\mathbf{x}}_S + \\ & [d_{Sy} + d_{Mx} \sin(2\theta_M) - d_{My} \cos(2\theta_M)] \hat{\mathbf{y}}_S + \\ & [d_{Sz} - d_{Mz}] \hat{\mathbf{z}} \end{aligned} \quad (437)$$

As in equation 427 and with the same sign conventions, the unit direction with respect to the monochromator to sample nominal path is

$$\mathbf{u}_{ms} = \sin(\delta_{10}) \hat{\mathbf{z}} + \cos(\delta_{10}) [\sin(\gamma_{10}) \hat{\mathbf{y}}_S + \cos(\gamma_{10}) \hat{\mathbf{x}}_S] \quad (438)$$

so that

$$\sin(\delta_{10}) = \frac{d_{Sz} - d_{Mz}}{D_{MS}} \quad (439)$$

$$\sin(\gamma_{10}) = \frac{d_{Sy} + d_{Mx} \sin(2\theta_M) - d_{My} \cos(2\theta_M)}{D_{MS} \cos(\delta_{10})} \quad (440)$$

with

$$D_{MS}^2 = (L_S + d_{Sx} - d_{Mx} \cos(2\theta_M) - d_{My} \sin(2\theta_M))^2 + (d_{Sy} + d_{Mx} \sin(2\theta_M) - d_{My} \cos(2\theta_M))^2 + (d_{Sz} - d_{Mz})^2 \quad (441)$$

Similarly, find for the sample to analyzer path, the unit direction from the sample to the analyzer element position is

$$\mathbf{u}_{sa} = (\mathbf{r}_A + \mathbf{d}_A - \mathbf{d}_S) / D_{SA} \quad (442)$$

where $D_{SA} = |\mathbf{r}_A + \mathbf{d}_A - \mathbf{d}_S|$. Now to put $\mathbf{d}_S$ and $\mathbf{d}_A$ in the same coordinate system

$$\hat{\mathbf{x}}_S = \cos(2\theta_S) \hat{\mathbf{x}}_A - \sin(2\theta_S) \hat{\mathbf{y}}_A \quad (443)$$

$$\hat{\mathbf{y}}_S = \sin(2\theta_S) \hat{\mathbf{x}}_A + \cos(2\theta_S) \hat{\mathbf{y}}_A \quad (444)$$

so that

$$\begin{aligned} \mathbf{u}_{sa} D_{SA} = & [L_A + d_{Ax} - d_{Sx} \cos(2\theta_S) - d_{Sy} \sin(2\theta_S)] \hat{\mathbf{x}}_A + \\ & [d_{Ay} + d_{Sx} \sin(2\theta_S) - d_{Sy} \cos(2\theta_S)] \hat{\mathbf{y}}_A + \\ & [d_{Az} - d_{Sz}] \hat{\mathbf{z}} \end{aligned} \quad (445)$$

As in equation 427 and with the same sign conventions, the unit direction with respect to the sample to analyzer nominal path is

$$\mathbf{u}_{sa} = \sin(\delta_{20}) \hat{\mathbf{z}} + \cos(\delta_{20}) [\sin(\gamma_{20}) \hat{\mathbf{y}}_A + \cos(\gamma_{20}) \hat{\mathbf{x}}_A] \quad (446)$$

so that

$$\sin(\delta_{20}) = \frac{d_{Az} - d_{Sz}}{D_{SA}} \quad (447)$$

$$\sin(\gamma_{20}) = \frac{d_{Ay} + d_{Sx} \sin(2\theta_S) - d_{Sy} \cos(2\theta_S)}{D_{SA} \cos(\delta_{20})} \quad (448)$$

with

$$D_{SA}^2 = (L_A + d_{Ax} - d_{Sx} \cos(2\theta_S) - d_{Sy} \sin(2\theta_S))^2 + (d_{Ay} + d_{Sx} \sin(2\theta_S) - d_{Sy} \cos(2\theta_S))^2 + (d_{Az} - d_{Sz})^2 \quad (449)$$

Finally for the analyzer to detector path, the unit direction from the analyzer to the detector position is

$$\mathbf{u}_{ad} = (\mathbf{r}_D - \mathbf{d}_A) / D_{AD} \quad (450)$$

where $D_{AD} = |\mathbf{r}_D - \mathbf{d}_A|$. Now to put $\mathbf{r}_D$ and $\mathbf{d}_A$ in the same coordinate system

$$\hat{\mathbf{x}}_A = \cos(2\theta_A) \hat{\mathbf{x}}_D - \sin(2\theta_A) \hat{\mathbf{y}}_D \quad (451)$$

$$\hat{\mathbf{y}}_A = \sin(2\theta_A) \hat{\mathbf{x}}_D + \cos(2\theta_A) \hat{\mathbf{y}}_D \quad (452)$$

so that

$$\begin{aligned} \mathbf{u}_{ad} D_{AD} = & [L_D - d_{Ax} \cos(2\theta_A) - d_{Ay} \sin(2\theta_A)] \hat{\mathbf{x}}_D + \\ & [d_{Ax} \sin(2\theta_A) - d_{Ay} \cos(2\theta_A)] \hat{\mathbf{y}}_D + \\ & [-d_{Az}] \hat{\mathbf{z}} \end{aligned} \quad (453)$$

As in equation 427 and with the same sign conventions, the unit direction with respect to the analyzer to detector nominal path is

$$\mathbf{u}_{ad} = \sin(\delta_{30}) \hat{\mathbf{z}} + \cos(\delta_{30}) [\sin(\gamma_{30}) \hat{\mathbf{y}}_D + \cos(\gamma_{30}) \hat{\mathbf{x}}_D] \quad (454)$$

so that

$$\sin(\delta_{30}) = \frac{-d_{Az}}{D_{AD}} \quad (455)$$

$$\sin(\gamma_{30}) = \frac{d_{Ax} \sin(2\theta_A) - d_{Ay} \cos(2\theta_A)}{D_{AD} \cos(\delta_{30})} \quad (456)$$

with

$$D_{AD}^2 = (L_D - d_{Ax} \cos(2\theta_A) - d_{Ay} \sin(2\theta_A))^2 + (d_{Ax} \sin(2\theta_A) - d_{Ay} \cos(2\theta_A))^2 + (-d_{Az})^2 \quad (457)$$

Crystal elements are assumed to be parallelopipeds. Then the element size collimation effect can be calculated by taking the individual crystal setting angle as measured from the straight through direction and so the reflection is from a face of the parallelopiped. It is also assumed that just two dimensions will characterize the reactor, sample and detector, those dimensions being a height and a width. The crystal elements also can have a thickness perpendicular to the reflection face. Thus the effective element size collimations can be approximated by

$$\alpha_{RM} = \frac{1}{2} [w_M \sin(\omega_{MR}) + t_M \cos(\omega_{MR}) + w_R] / D_{RM} \quad (458)$$

$$\alpha_{MS} = \frac{1}{2} [w_M \sin(\omega_{MS}) + t_M \cos(\omega_{MS}) + w_S] / D_{MS} \quad (459)$$

$$\alpha_{SA} = \frac{1}{2} [w_A \sin(\omega_{AS}) + t_A \cos(\omega_{AS}) + w_S] / D_{SA} \quad (460)$$

$$\alpha_{AD} = \frac{1}{2} [w_A \sin(\omega_{AD}) + t_A \cos(\omega_{AD}) + w_D] / D_{AD} \quad (461)$$

where ω_{MR} and ω_{MS} are the effective monochromator crystal offset angles as seen from the reactor and sample respectively, and similarly for ω_{AD} and ω_{AS} . These effective offset angles are

$$\omega_{MR} = \omega_m = |\omega_{mi} - \gamma_{00}| \quad (462)$$

$$\omega_{MS} = |2\theta_M - \omega_{mi} + \gamma_{10}| \quad (463)$$

$$\omega_{AS} = \omega_a = |\omega_{ai} - \gamma_{20}| \quad (464)$$

$$\omega_{AD} = |2\theta_A - \omega_{ai} + \gamma_{30}| \quad (465)$$

where ω_{mi} and ω_{ai} are the monochromator and analyzer offset angles with respect to the original reactor to monochromator line and sample to analyzer line respectively.

For the vertical effective crystal size collimations we use

$$\beta_{RM} = \frac{1}{2} [h_M + h_R] / D_{RM} \quad (466)$$

$$\beta_{MS} = \frac{1}{2} [h_M + h_S] / D_{MS} \quad (467)$$

$$\beta_{SA} = \frac{1}{2} [h_A + h_S] / D_{SA} \quad (468)$$

$$\beta_{AD} = \frac{1}{2} [h_A + h_D] / D_{AD} \quad (469)$$

References

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[CandA] Chessser, N.J. and Axe, J.D., Acta Cryst. 29, 160(1973).